Leibniz Universität Hannover

Laboratory AI-ready scientific knowledge

Markus Stocker



Welcome and introductions

- Second year for this laboratory
- Exciting opportunity to transfer our own research into teaching
- About you and about me
  - What are you studying, in which institute and faculty
  - What motivated you to join this laboratory
  - What do you expect to learn

What is a laboratory?

- Learning by doing
- Learn new methods and skills and master them by applying them
- Only a few introductory lectures, the rest is hands-on
- Get things done in class, no homework
- No additional exams, essays, assignments, or other tasks
- This, however, means that attendance in class is required
- Inform if you can’t attend due to for instance sickness

This laboratory in a nutshell (I)

- Learn how scientific findings can be produced as machine-readable data
- Learn how systems can manage and provide access to such data
- Learn how systems can assist the reuse of scientific knowledge
- Master the methods by applying them to selected research articles
- Gain expertise in using systems that support you in knowledge reuse

This laboratory in a nutshell (II)

- Along the way, you will understand what the problem is
- As well as why the problem matters
- What others have been doing about the problem
- How we at TIB (and LUH) have been approaching the problem
- What remains to be done and how the future could look like
- By processing code, you'll also learn disciplinary data analysis techniques!

This laboratory in a nutshell (III)

- Learn about the TIB Knowledge Loom (knowledgeloom.tib.eu)
- Production, curation, publication, use of machine-readable scientific knowledge
- Learn a method that ensures scientific knowledge is produced reusable
- Learn how the TIB Knowledge Loom supports knowledge reuse
- How it is rooted in the Open Research Knowledge Graph (ORKG, orkg.org)

This laboratory in a nutshell (IV)

- Apply the Loom method on selected articles
- This can be done individually or together with a fellow student
- You will also have the opportunity to present your work to others in class
- Ideally, the articles are authored by researchers at your institute or faculty

General remarks (I)

- This is an interdisciplinary laboratory: students from all disciplines are welcome
- The cap is at approximately 20 participants
- Priority is given to **Master** students, but doctoral students are welcome, too
- Number of credits depends on your degree programme (3/5-6 ECTS)
- The expected hours also depend on your degree programme (2-4 hours/week)
- The laboratory is not graded, you either pass or fail
- To pass you'll need to show that you master the learned methods

General remarks (II)

- Bring your laptop to classes, you'll need it for most of the semester
- Also useful are the power cord as well as an external mouse
- You should be able to read and understand articles (in your field of science)
- Also useful are basic skills in R and/or Python scripting (code reading)
- In some disciplines, understanding of statistical data analysis may be a plus
- In computer science, understanding of algorithm/model evaluation is useful
- A curiosity for research data stewardship rounds up the profile
- Make sure to register for the laboratory in QIS!

Schedule

Part	Day	
I	13.04.	Lectures (marked red means cancelled)
	20.04.	
	27.04.	
	04.05.	
II	11.05.	Laboratory work (marked yellow may mean cancelled; marked green is planned for presentations)
	18.05.	
	01.06.	
	08.06.	
	15.06.	
	22.06.	
	29.06.	
	06.07.	
	13.07.	

Part I: Lectures

- Introduce the problems, challenges, motivation, research
- Introduction to research data & scientific knowledge management
- Introduction to solutions, approaches, tools, infrastructure, and services

Part II: Laboratory work

- Primary task is to turn original articles into *Loom articles*
- This is achieved by applying the Loom method you'll learn in Part I
- Each team (1-2 students) must process at least one article
- Teams may identify articles by researchers at their respective institutes
- I will also propose a set of candidate articles
- Hands-on work with a computer, research data and code, supervised by me
- There will be time to present individual results in class (see schedule)

General remarks (I)

- Weekly up to 4 hours (4 x 45 minutes = 180 minutes or 3 hours)
- The room is reserved from 2 - 6 pm (4 hours)
- The weekly sessions are structured as follows

Part A	**2:00** - 3:30 pm	90 min
Break	3:30 - 4:00 pm	30 min
Part B	4:00 - **5:30** pm	90 min

General remarks (II)

- Degree programmes in which the module is 4 hours/week
  - Faculty of Electrical Engineering and Computer Science
  - You join weekly for both Part A and Part B (2:00 - 5:30 pm)
- Degree programmes in which the module is 2 hours/week
  - Economics and Management
  - For Part I (lectures) you'll need to join for both Part A **and** Part B!
  - For Part II (laboratory work) you can choose to join either for Part A **or** Part B
  - You can skip two sessions of your choice in Part II to compensate overtime in Part I
  - Form a larger team (3-4 students) or apply the Loom method on fewer results
- Participants without ECTS requirements (e.g., doctoral students)
  - Join us for Part I (lectures) to learn the Loom method
  - Feel free to join or skip Part II, apply the method in your own work

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Questions

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Problem



failure than in LV tissue samples from unused donor hearts (*Figure 1A*). As shown by electrophoretic mobility shift assays, IRE binding activity was significantly reduced in failing hearts (most pronounced in patients with ischemic cardiomyopathy) (*Figure 1B*). Protein expression levels of the transferrin receptor were significantly lower in failing hearts than in the controls (*Figure 1C*).

Targeted Irp deletion in mice induces ID in the myocardium

We generated mice with a cardiomyocyte-targeted deletion of *Irp1* and *Irp2* (Cre-Irp1/2$^{0/f}$) to address Irp function in the heart (*Figure 2A*). Cre-Irp1/2$^{0/f}$ mice were born at the expected Mendelian inheritance ratio and survived into adulthood. Reverse transcriptase polymerase chain reaction on LV myocardium and isolated cardiomyocytes demonstrated near-complete Cre-mediated deletion of Irp1 and Irp2 mRNAs in cardiomyocytes from Cre-Irp1/2$^{f/f}$ mice compared with littermates lacking the Cre transgene (Irp1/2$^{f/f}$) (*Figure 2B*). Irp1 and Irp2 protein expression was markedly reduced in LV myocardium and barely detectable in isolated cardiomyocytes from Cre-Irp1/2$^{0/f}$ mice (*Figure 2C and D*). Irp1 and Irp2 protein expression in the liver was similar in Cre-Irp1/2$^{f/f}$ and Irp1/2$^{f/f}$ mice (*Figure 2C and D*). IRE binding activity was strongly reduced in isolated cardiomyocytes from Cre-Irp1/2$^{0/f}$ mice (*Figure 2E*), confirming near-complete Cre-mediated recombination. Iron-regulatory protein/IRE-regulated proteins involved in iron transport and storage were differentially regulated in cardiomyocytes from Cre-Irp1/2$^{0/f}$ mice: the transferrin receptor was down-regulated (25 ± 14% of Irp1/2$^{f/f}$ controls, $P = 0.006$), whereas ferroportin (325 ± 9%, $P = 0.003$) and ferritin H-chain (249 ± 35%, $P = 0.012$) were up-regulated ($n = 3$ per group; representative immunoblots are presented in *Figure 2F*). As a result, iron concentration in cardiomyocytes was significantly reduced in Cre-Irp1/2$^{0/f}$ mice (*Figure 2G*). Likewise, iron concentration in the left ventricle was reduced in Cre-Irp1/2$^{0/f}$ mice compared with Irp1/2$^{f/f}$ controls, whereas iron concentrations in the M. quadriceps femoris and liver were not affected (*Figure 2H*). Iron concentration in the left ventricle was normal in Cre mice showing that cardiac ID in Cre-Irp1/2$^{0/f}$ mice was not related to Cre transgene expression per se (*Figure 2H*). Haem and myoglobin concentrations were significantly reduced in the left ventricle of Cre-Irp1/2$^{0/f}$ mice (*Figure 2I and J*). Copper and free radical concentrations in the left ventricle were similar in Cre-Irp1/2$^{f/f}$ and Irp1/2$^{0/f}$ mice (see Supplementary material online, *Figure S2*).

Cre-Irp1/2$^{0/f}$ mice did not show an obvious phenotype under baseline conditions. Body mass, heart mass, LV mass, and cardiomyocyte cross-sectional area were similar in Cre-Irp1/2$^{f/f}$ and Irp1/2$^{0/f}$ mice under baseline conditions (see Supplementary material online, *Table S1*). On echocardiography, LV end-diastolic and end-systolic dimensions and LV systolic and diastolic function were similar in both genotypes (see Supplementary material online, *Table S1*). Cre-Irp1/2$^{0/f}$ mice were not anaemic and had a normal peripheral blood count (see Supplementary material online, *Table S2*).

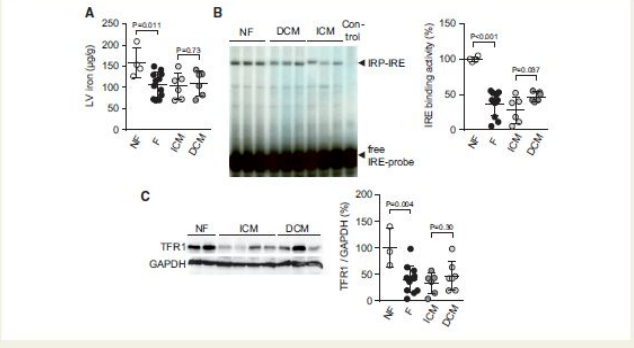


Figure 1 *Reduced IRP activity and iron content in failing human hearts.* (*A*) Non-haem iron concentration in left ventricular (LV) tissue samples from non-failing donors (NF) and patients with cardiac failure (*F*) due to ischemic cardiomyopathy (ICM) or dilated cardiomyopathy (DCM); $n = 4$–6 per group. (*B*) Representative electrophoretic mobility shift assay and summary data showing iron-responsive element (IRE) binding activity in LV tissue samples; $n = 4$–6 (control, no sample loaded). (*C*) Representative immunoblot and summary data showing transferrin receptor 1 (TFR1) and GAPDH protein expression in LV tissue samples; $n = 3$–7. *P* values were determined by two independent sample *t*-test.

https://doi.org/10.1093/eurheartj/ehw333

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Scientific knowledge is data

Data expressed in scientific articles

This data is not machine readable

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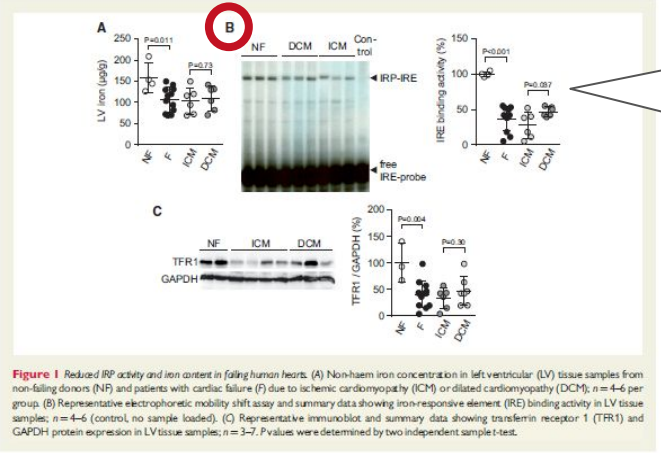
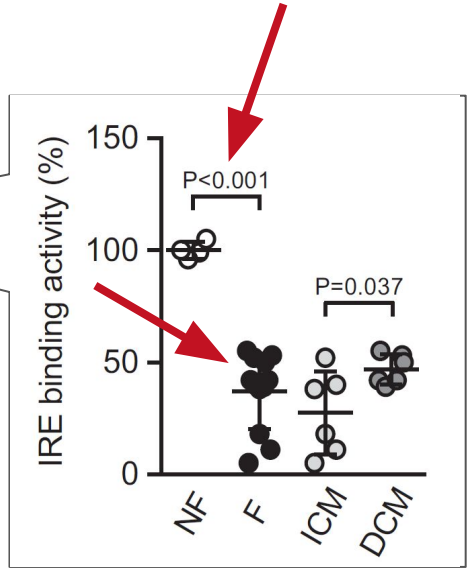


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Knowledge reuse
relies on manual knowledge extraction

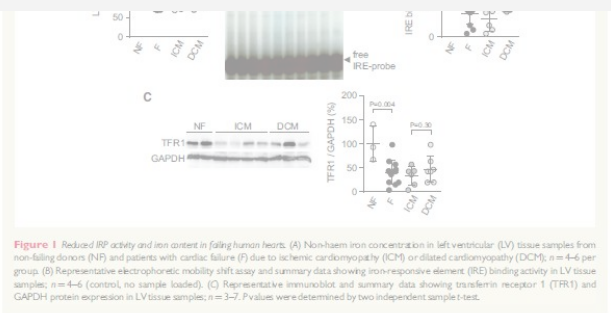
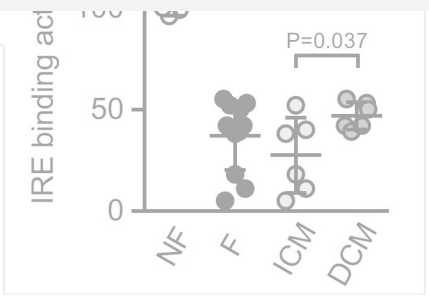


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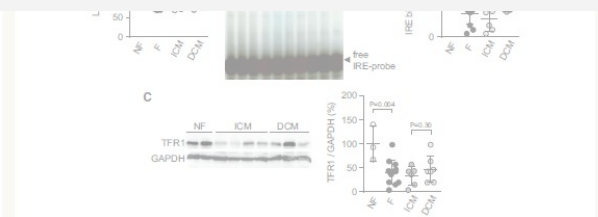
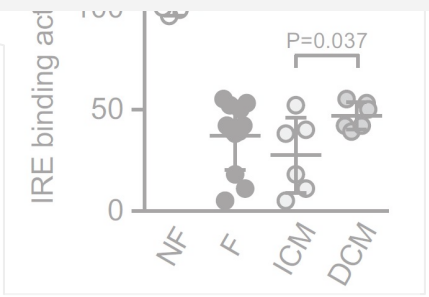


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Manual knowledge extraction is expensive, inaccurate, hard to organize and scale



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Questions

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Data and knowledge

What are data?

		case fatality rate	average basic reprod...	primary mode of tran...
1	grey = not plotted	healthy adult	also known as "R0"	
2	Anthrax (untreated)	20	6.5	airborne
3	Bird Flu (H5N1)	60	1	airborne droplet
4	Bubonic Plague (untr...	60	1	bites
5	C.Difficile	24	1.25	fecal-oral
6	Campylobacter	1.20	0.19	food
7	Chicken Pox	0.00	8.5	airborne droplet
8	Cholera	1.63	2.13	
9	Dengue Fever	5	3	bites
10	Diphtheria			
11	E.coli			
12	Ebola			
13	Hand, Foot and Mouth			





What is knowledge?

- Knowledge as **accounted** semantic information
- **Semantic information** as truthful, meaningful, and well-formed **data**
- Example
  - Pythagoras theorem $a^2 + b^2 = c^2$ is a piece of semantic information; it informs *that*
  - A mathematician can prove the theorem with multiple proofs
  - Hence, a mathematician knows *why* Pythagoras theorem is in fact true
  - The mathematician can explain / justify / provide an account for the piece of semantic information

Floridi, L. (2005). Is Semantic Information Meaningful Data? In Philosophy and Phenomenological Research (Vol. 70, Issue 2, pp. 351–370). Wiley. https://doi.org/10.1111/j.1933-1592.2005.tb00531.x

What is *scientific* knowledge?

- **Systematically** accounted semantic information
- It is the type of knowledge of primary interest in this laboratory
- It is knowledge expressed in the scientific literature
- Hence, our interest is for
  - The truthful, meaningful, and well-formed data
  - That constitute the semantic information
  - Systematically accounted for in scientific knowledge
  - Expressed in academic articles

What are data?

		case fatality rate	average
1	grey = not plotted	healthy adult	also
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5	C.Difficile	24	
6	Campylobacter	1.20	
7	Chicken Pox	0.00	
8	Cholera	1.63	
9	Dengue Fever	5	
10	Diphtheria	7.50	
11	E.coli	4	
12	Ebola	50	
13	Hand, Foot and Mouth	0.06	



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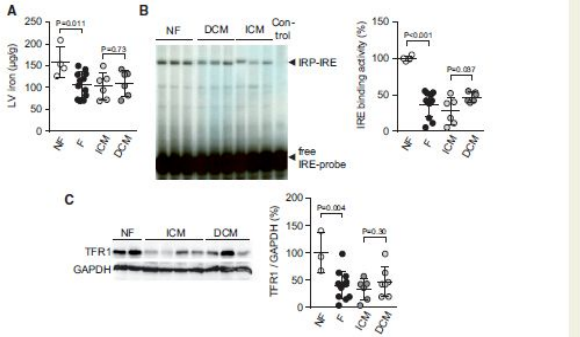


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Data literacy

Data literacy

- The ability to read, understand, create, and communicate data as information
- The competencies involved in working with data
- Understand what data mean, draw conclusions from data
- Data analysis needed in the research, business, government and society
- Some level of data literacy is important for many, if not most

Data literacy toolbox

- Find (digital) data
- Access and read data
- Process and analyse data
- Manage and publish data

Find data

- Finding data is comparable to finding any other digital artefact, say a video
- It is as difficult to find *relevant* data as it is (nowadays) to find anything digital
- Finding data is supported by
  - Standardised metadata
  - Domain specific and general purpose data repositories
- Databases support data search comparable to fulltext
- This is, however, only supported by the most advanced systems
- Data repositories are typically limited to search on metadata



Access data

- Data and service specific
- Data can be open, closed, embargoed, lost, available (upon request), etc.
- Today, access is mostly over the Web, but not necessarily openly
- Multitude of data exchange protocols (HTTP, FTP widely in use)
- Access may or may not need human interaction (APIs)
- In direct exchange, data may also be shared by other (electronic) means



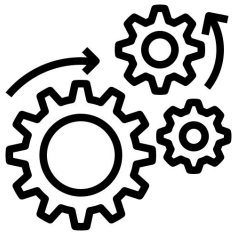
Read data

- Data (and service) specific
- Textual data (say, CSV) is widely supported, in tools and scripting languages
- Much data is binary, in both open (say, PNG images) and closed formats
- Open binary formats also well supported in tools and scripting languages
- For closed binary formats you typically rely on commercial software



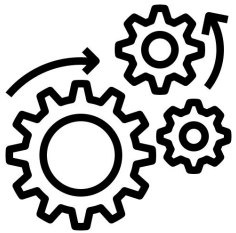
Process and analyse data (I)

- (Research) context-specific
- How you need to process and analyse data depends on your (research) aim
- The toolbox of methods is enormous: from descriptive stats to deep learning
- Many computational environments and scripting languages
- A lot of time is spent on data processing (80%) and too little on analysis (20%)
- Preparing data for analysis is unfortunately very resource intensive



Process and analyse data (II)

- Mostly on local computers with locally-installed software (say, MATLAB)
- Meanwhile also cloud-based systems exist, Jupyter probably most prominent
- And of course there is also HPC, which is an own world
- Cloud-based systems have many advantages, e.g. shared environment



Data management

- Data consumed and produced should be managed
- How, very much depends on the complexity of the research project
- For a doctoral student working alone, management could be simple
- For a postdoc in a large consortium, management will be more complex
- Simple strategy may just involve consistent file naming and directory structure
- More complex strategies will rely on data management planning and tooling

Data publication

- Data repositories play a major role in data access and archival
- They support meeting best practices, some with expert curation
- As a researcher, make sure to submit your data to a trusted repository
- You do this so that **your own future self** will have access to your data
- Repository may be
  - Domain specific, e.g. PANGAEA in Earth Science (https://pangaea.de)
  - Institutional, e.g. LUH Research Data Repository (https://data.uni-hannover.de/)
  - National, e.g. Nationale Forschungsdaten Infrastruktur (https://nfdi.de)
  - General purpose, e.g. Zenodo (https://zenodo.org)

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Search for “temperature” at pangaea.de

Code management and publication

- Given its importance, code must be managed as well
- Strategy to support collaboration and versioning of code
- As for data, ensure a release (e.g. at article publication time) is deposited
- GitHub-Zenodo integration makes such deposition easy





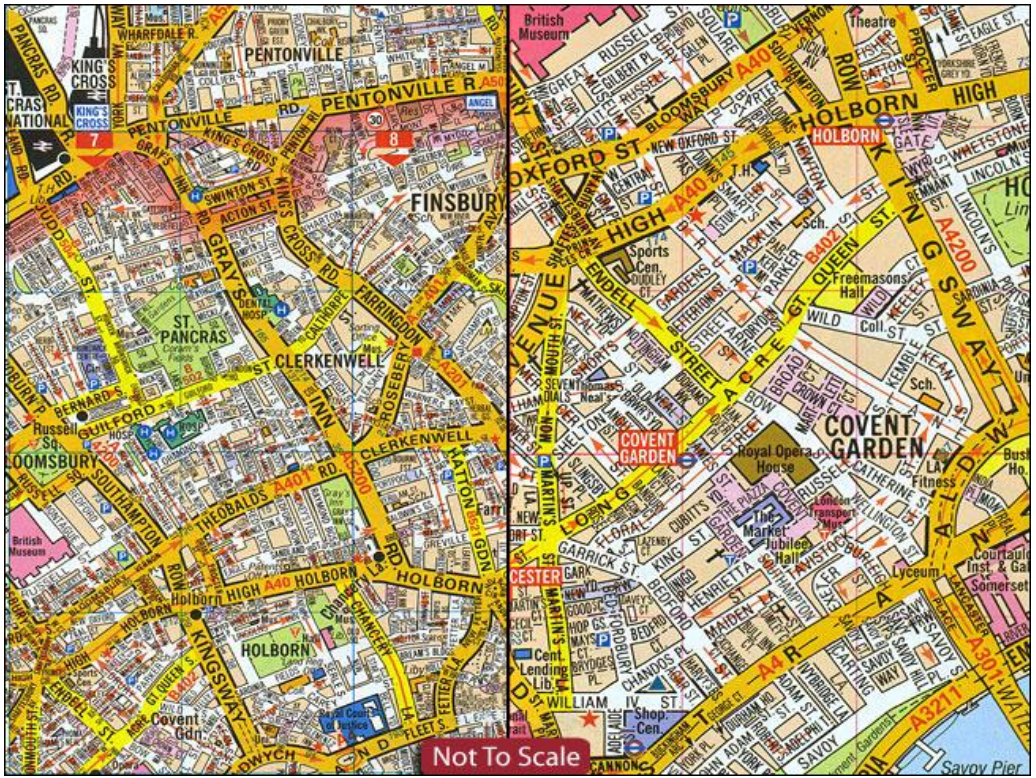
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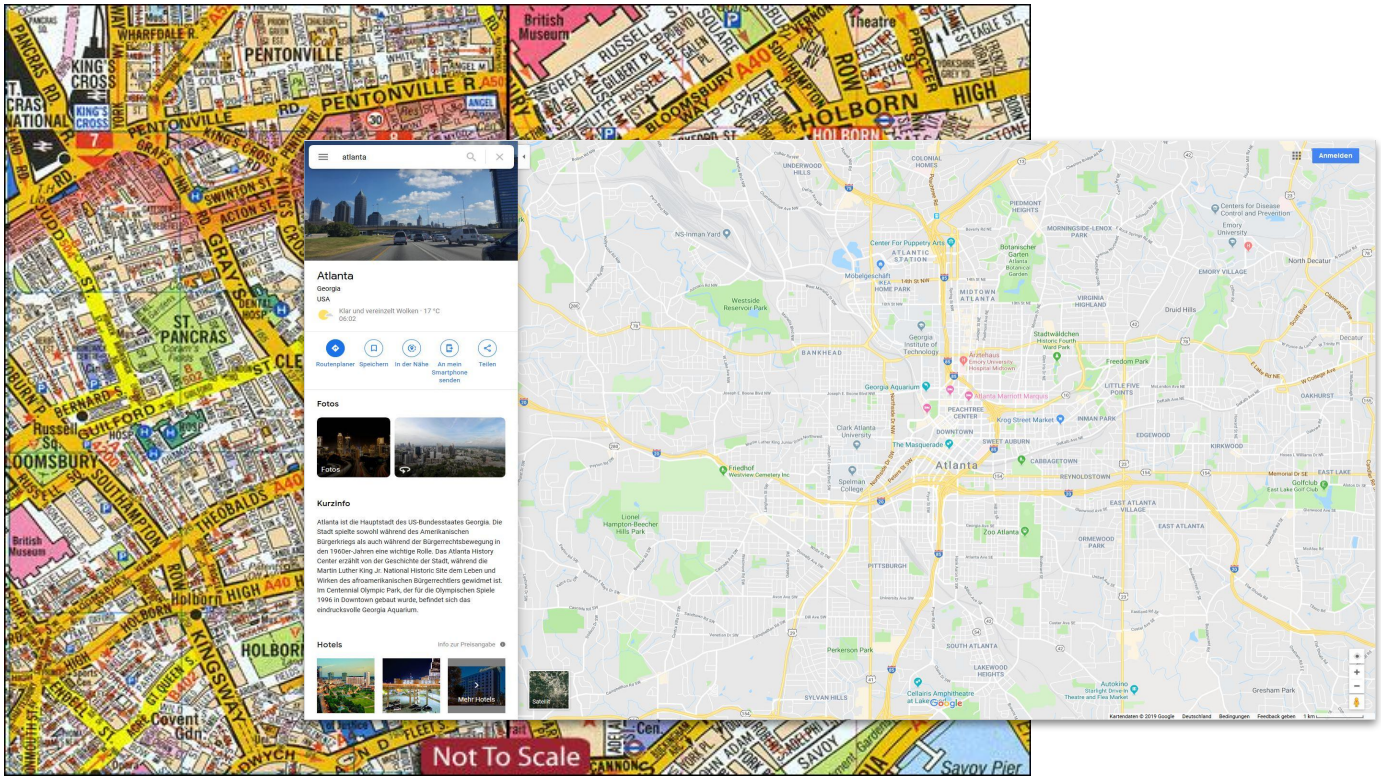
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State of the art

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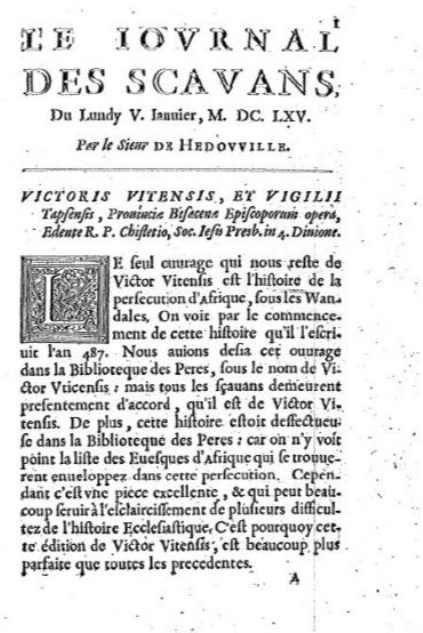


Digitalization in the 21st century



Digitalization in research

1665



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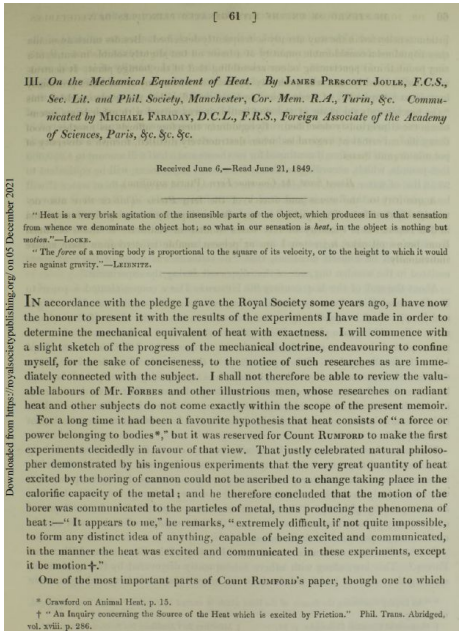
VICTORIS VITENSIS, ET VIGILII Tapſenſis, Prouinciæ Biſcenæ Epiſcoporum opera, Edente R. P. Chifletio, Soc. Ieſu Presb. in 4. Dinione.

LE ſeul ouurage qui nous reſte de Victor Vitenſis eſt l'hiſtoire de la perſecution d'afrique, ſous les Wandales. On voit par le commencement de cette hiſtoire qu'il l'eſcriuit l'an 487. Nous auions deſia cet ouurage dans la Biblioteque des Peres, ſous le nom de Victor Vticenſis : mais tous les ſçauans demeurent preſentement d'accord, qu'il eſt de Victor Vitenſis. De plus, cette hiſtoire eſtoit deffectueuſe dans la Biblioteque des Peres : car on n'y voit point la liſte des Eueſques d'afrique qui ſe trouuerent enueloppez dans cette perſecution. Cependant c'eſt vne piece excellente, & qui peut beaucoup ſeruir à l'eſclairciſſement de pluſieurs difficultez de l'hiſtoire Eccleſiaſtique, C'eſt pourquoy cette édition de Victor Vitenſis, eſt beaucoup plus parfaite que toutes les precedentes.

A

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1850



[61]

III. *On the Mechanical Equivalent of Heat. By* JAMES PRESCOTT JOULE, *F.C.S., Sec. Lit. and Phil. Society, Manchester, Cor. Mem. R.A., Turin, &c. Communicated by* MICHAEL FARADAY, *D.C.L., F.R.S., Foreign Associate of the Academy of Sciences, Paris, &c. &c. &c.*

Received June 6.—Read June 21, 1849.

"Heat is a very brisk agitation of the insensible parts of the object, which produces in us that sensation from whence we denominate the object hot; so what in our sensation is *heat*, in the object is nothing but *motion*."—LOCKE.

"The *force* of a moving body is proportional to the square of its velocity, or to the height to which it would rise against gravity."—LEIBNITZ.

IN accordance with the pledge I gave the Royal Society some years ago, I have now the honour to present it with the results of the experiments I have made in order to determine the mechanical equivalent of heat with exactness. I will commence with a slight sketch of the progress of the mechanical doctrine, endeavouring to confine myself, for the sake of conciseness, to the notice of such researches as are immediately connected with the subject. I shall not therefore be able to review the valuable labours of Mr. FORBES and other illustrious men, whose researches on radiant heat and other subjects do not come exactly within the scope of the present memoir.

For a long time it had been a favourite hypothesis that heat consists of "a force or power belonging to bodies*," but it was reserved for Count RUMFORD to make the first experiments decidedly in favour of that view. That justly celebrated natural philosopher demonstrated by his ingenious experiments that the very great quantity of heat excited by the boring of cannon could not be ascribed to a change taking place in the calorific capacity of the metal; and he therefore concluded that the motion of the borer was communicated to the particles of metal, thus producing the phenomena of heat:—"It appears to me," he remarks, "extremely difficult, if not quite impossible, to form any distinct idea of anything, capable of being excited and communicated, in the manner the heat was excited and communicated in these experiments, except it be motion†."

One of the most important parts of Count RUMFORD's paper, though one to which

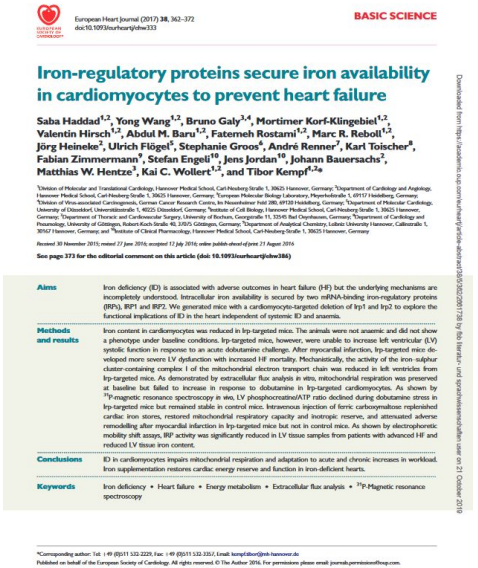
* Crawford on Animal Heat, p. 15.

† "An Inquiry concerning the Source of the Heat which is excited by Friction." Phil. Trans. Abridged, vol. xviii. p. 286.

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2017



European Heart Journal (2017) **38**, 362–372
doi:10.1093/eurheartj/ehw333

BASIC SCIENCE

Iron-regulatory proteins secure iron availability in cardiomyocytes to prevent heart failure

Saba Haddad[1,2], Yong Wang[1,2], Bruno Galy[3,4], Mortimer Korf-Klingebiel[1,2], Valentin Hirsch[1,2], Abdul M. Baru[1,2], Fatemeh Rostami[1,2], Marc R. Reboll[1,2], Jörg Heineke[2], Ulrich Flögel[5], Stephanie Groos[6], André Renner[7], Karl Toischer[8], Fabian Zimmermann[9], Stefan Engelli[10], Jens Jordan[10], Johann Bauersachs[2], Matthias W. Hentze[3], Kai C. Wollert[1,2], and Tibor Kempf[1,2]*

See page 373 for the editorial comment on this article (doi: 10.1093/eurheartj/ehw386)

Aims Iron deficiency (ID) is associated with adverse outcomes in heart failure (HF) but the underlying mechanisms are incompletely understood. Intracellular iron availability is secured by two mRNA-binding iron-regulatory proteins (IRPs), IRP1 and IRP2. We generated mice with a cardiomyocyte-targeted deletion of Irp1 and Irp2 to explore the functional implications of ID in the heart independent of systemic ID and anaemia.

Methods and results Iron content in cardiomyocytes was reduced in Irp-targeted mice. The animals were not anaemic and did not show a phenotype under baseline conditions. Irp-targeted mice, however, were unable to increase left ventricular (LV) systolic function in response to an acute dobutamine challenge. After myocardial infarction, Irp-targeted mice developed more severe LV dysfunction with increased HF mortality. Mechanistically, the activity of the iron–sulphur cluster-containing complex I of the mitochondrial electron transport chain was reduced in left ventricles from Irp-targeted mice. As demonstrated by extracellular flux analysis *in vitro*, mitochondrial respiration was preserved at baseline but failed to increase in response to dobutamine in Irp-targeted cardiomyocytes. As shown by ^{31}P-magnetic resonance spectroscopy *in vivo*, LV phosphocreatine/ATP ratio declined during dobutamine stress in Irp-targeted mice but remained stable in control mice. Intravenous injection of ferric carboxymaltose replenished cardiac iron stores, restored mitochondrial respiratory capacity and inotropic reserve, and attenuated adverse remodelling after myocardial infarction in Irp-targeted mice but not in control mice. As shown by electrophoretic mobility shift assays, IRP activity was significantly reduced in LV tissue samples from patients with advanced HF and reduced LV tissue iron content.

Conclusions ID in cardiomyocytes impairs mitochondrial respiration and adaptation to acute and chronic increases in workload. Iron supplementation restores cardiac energy reserve and function in iron-deficient hearts.

Keywords Iron deficiency • Heart failure • Energy metabolism • Extracellular flux analysis • ^{31}P-Magnetic resonance spectroscopy

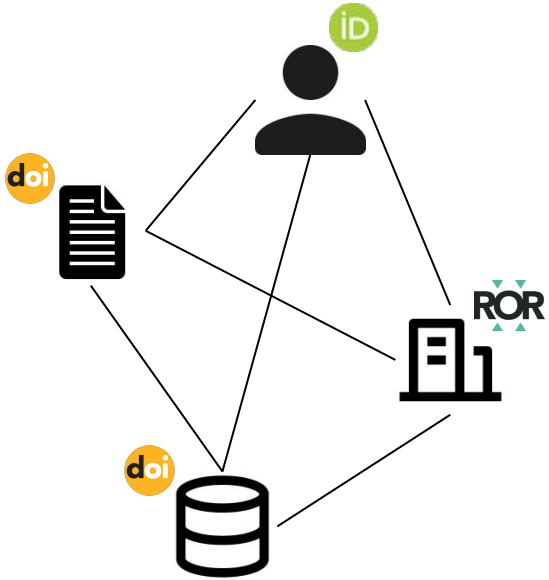
*Corresponding author: Tel: +49 (0)511 532-2229, Fax: +49 (0)511 532-2317, Email: kempf.tibor@mh-hannover.de

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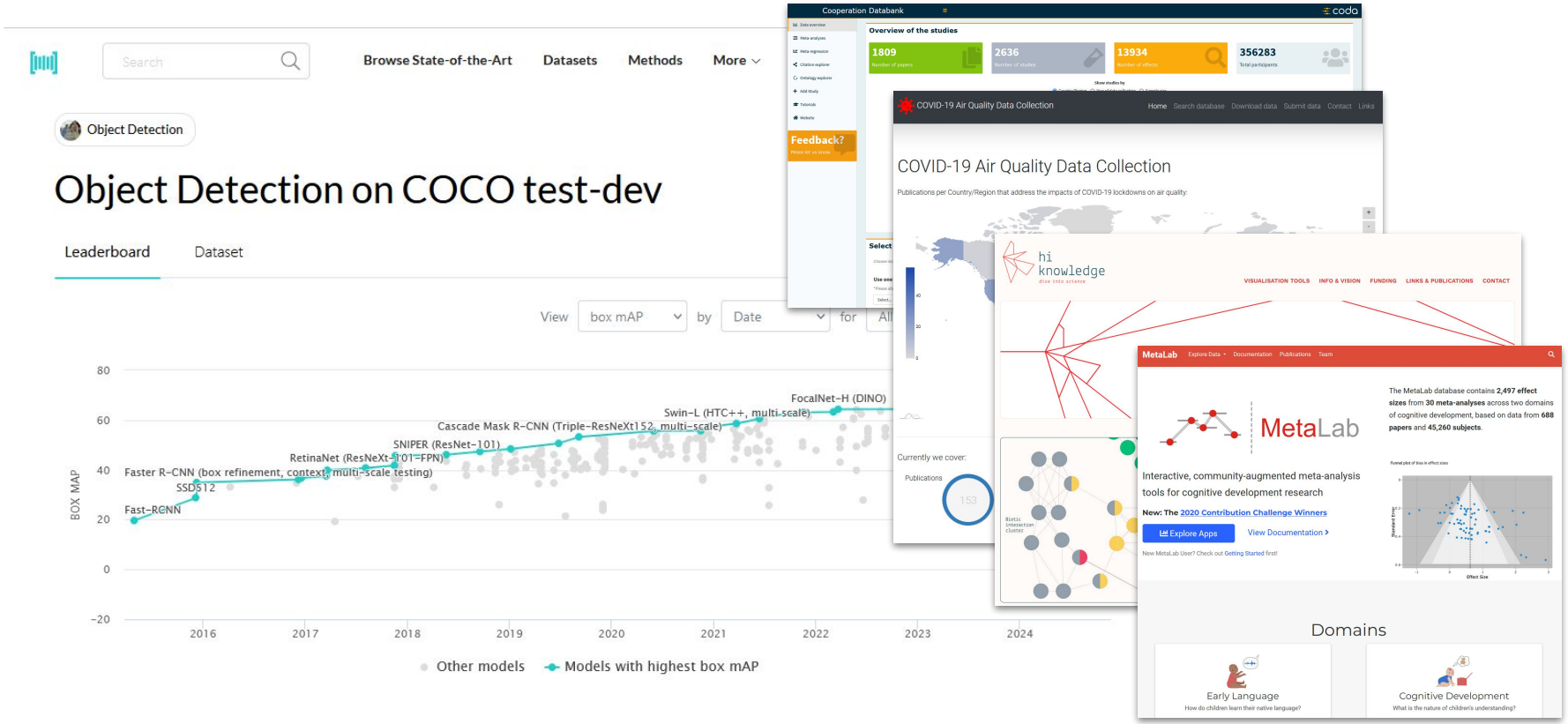
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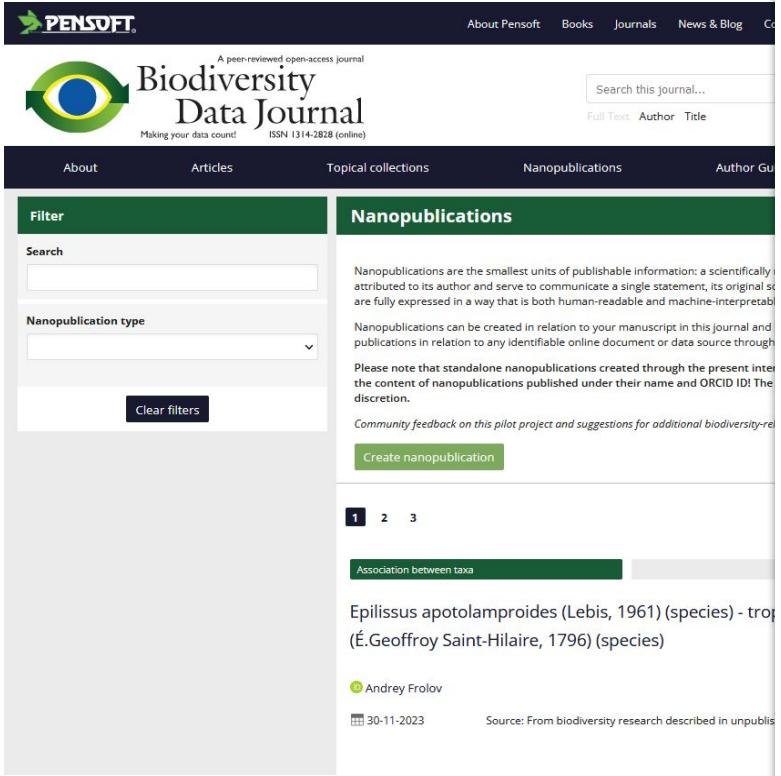
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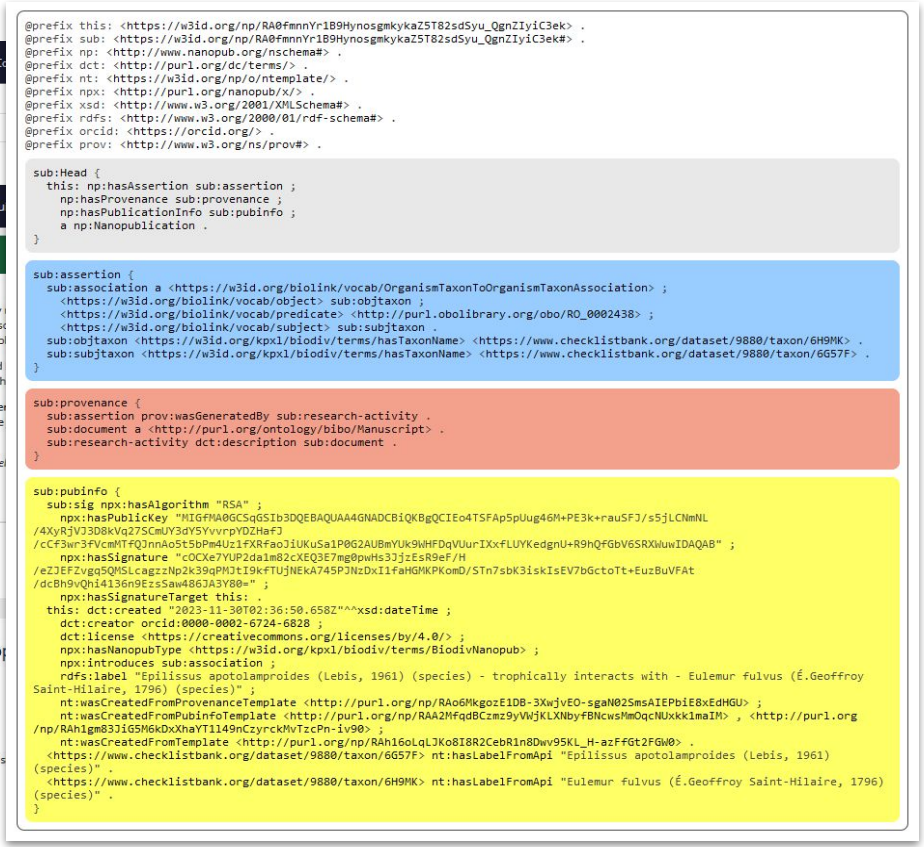
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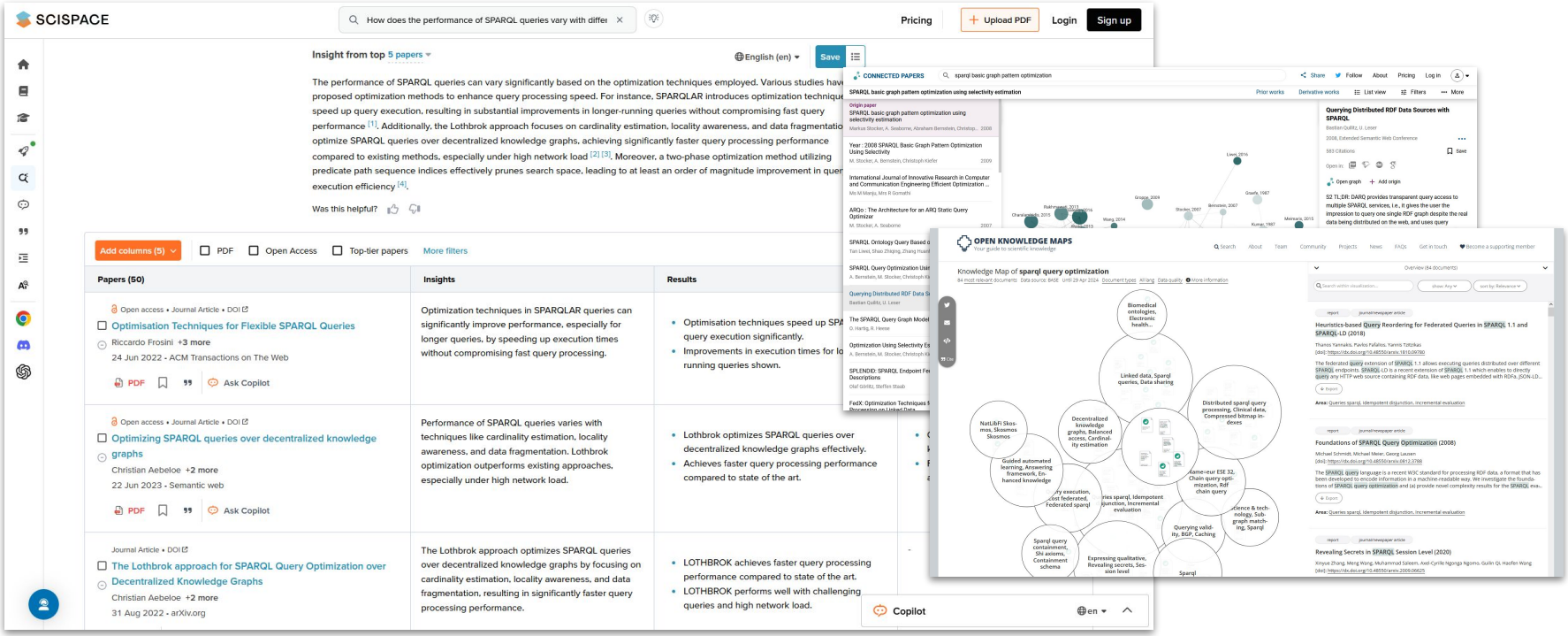
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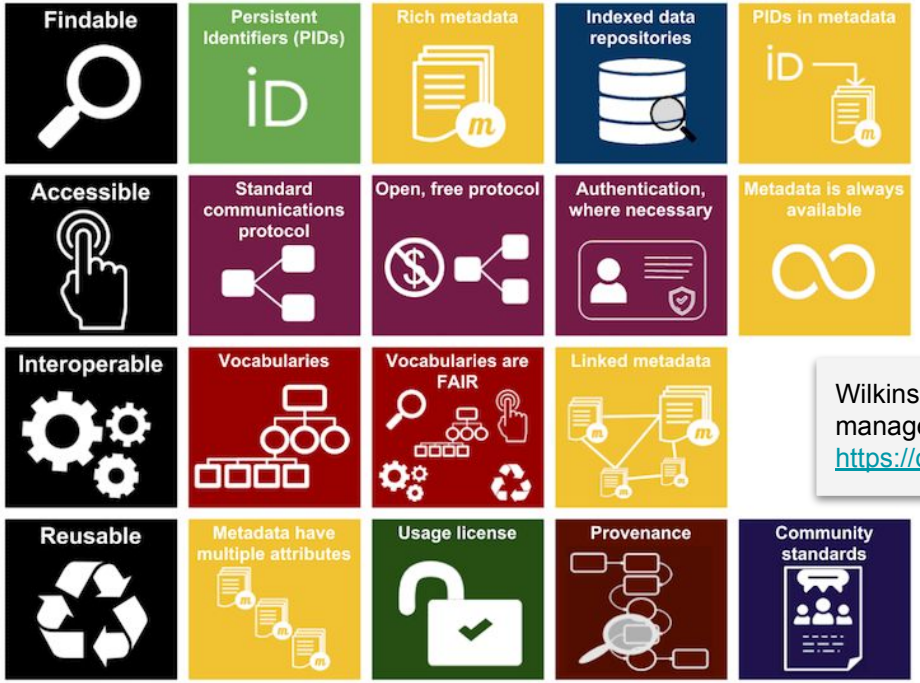
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Wilkinson et al. (2016). The FAIR Guiding Principles for scientific data management and stewardship. Scientific Data. https://doi.org/10.1038/sdata.2016.18

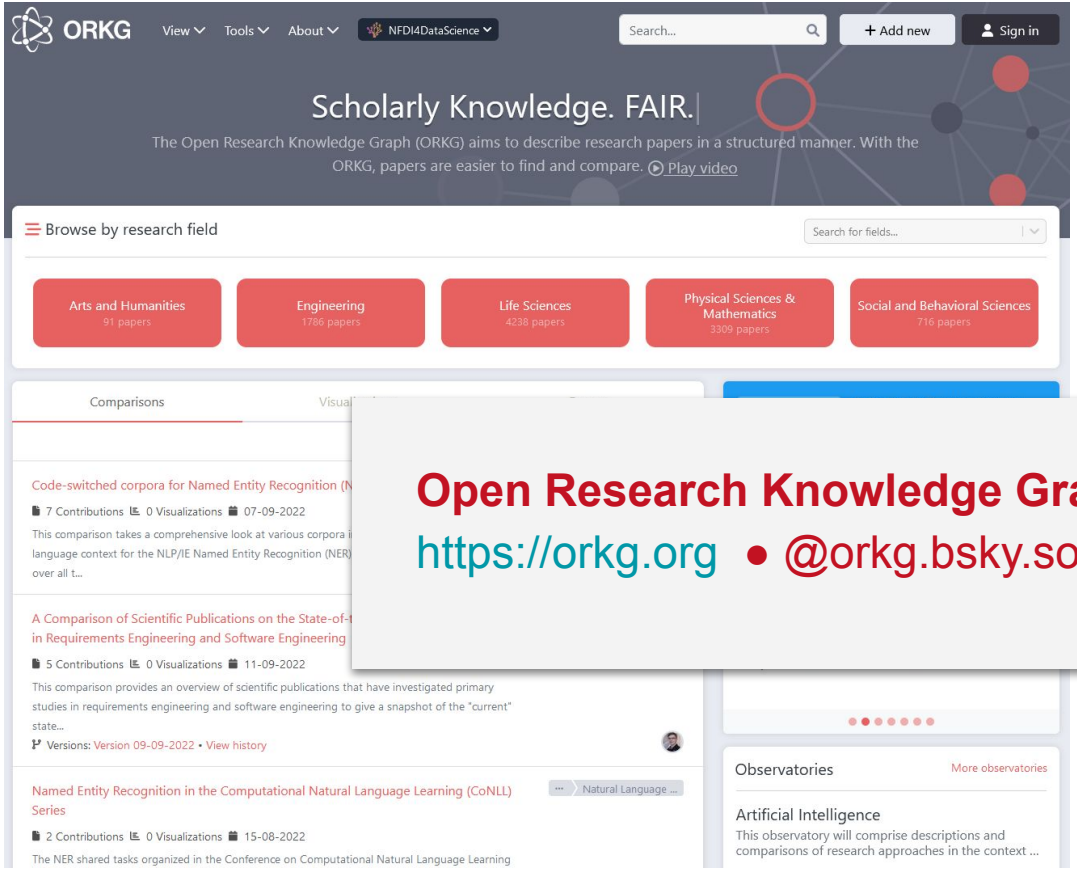
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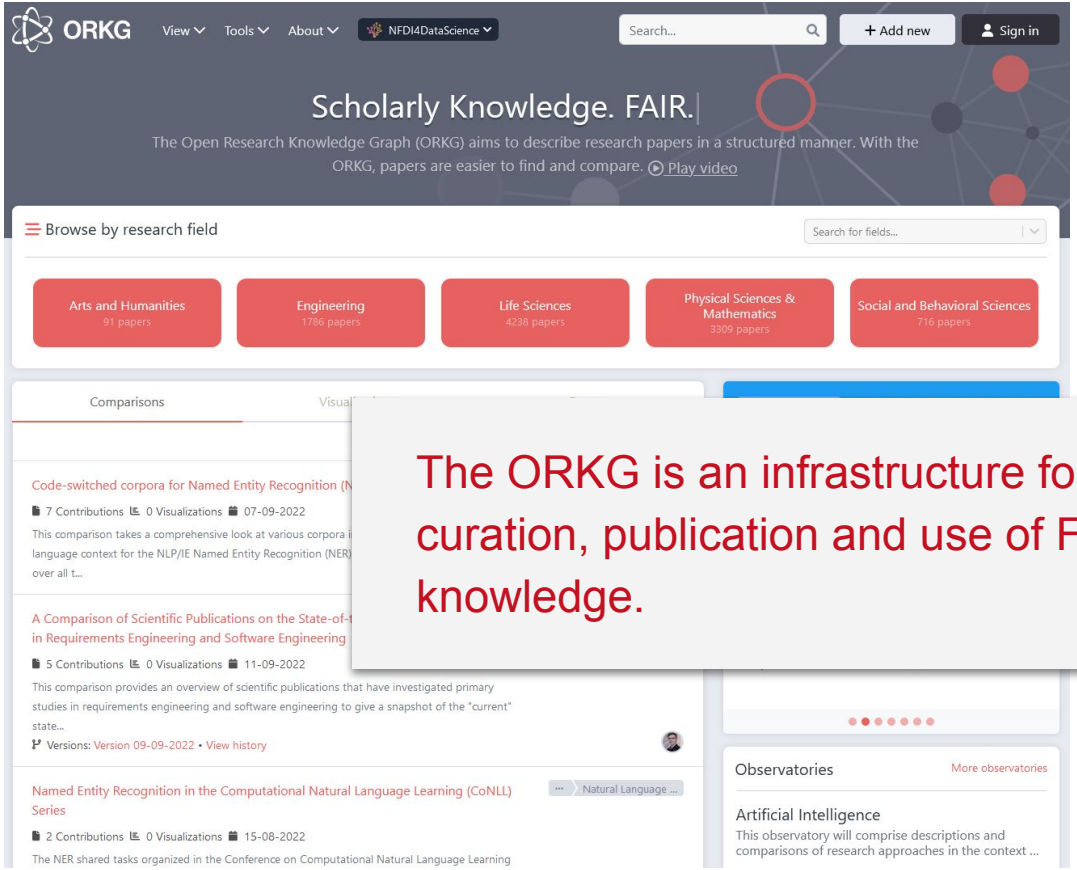
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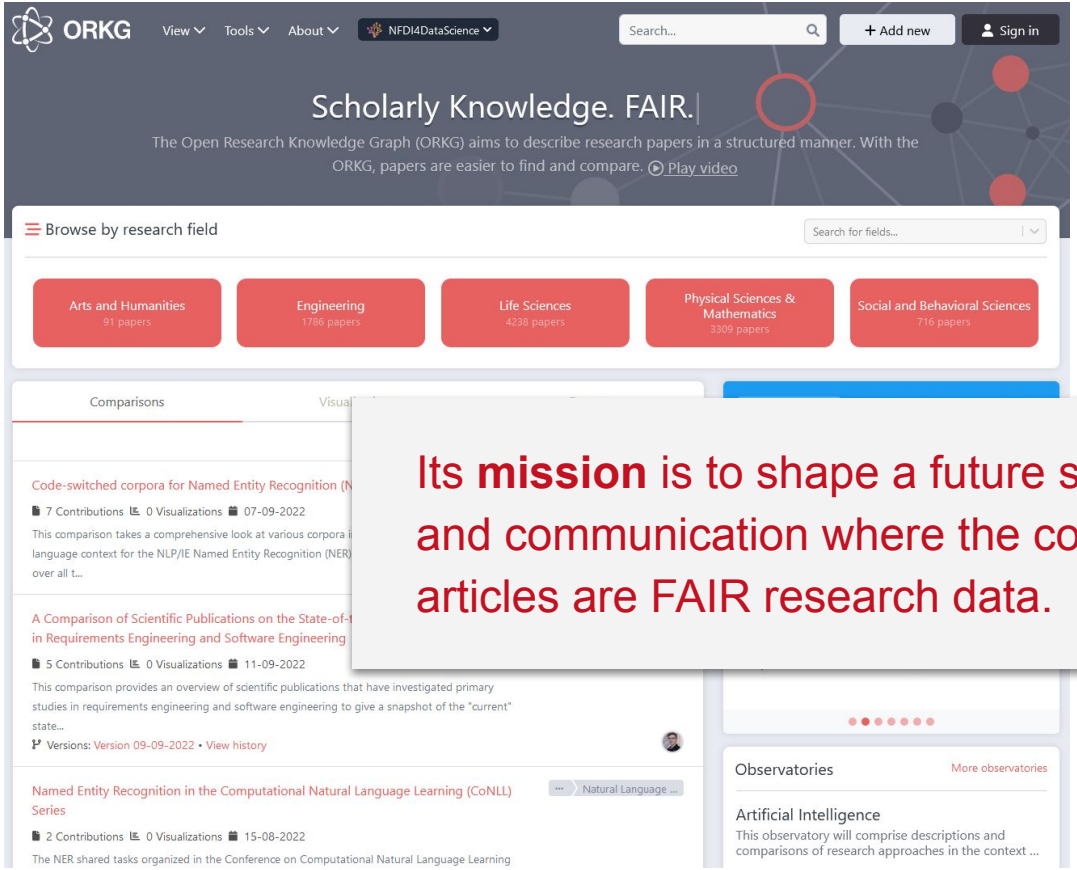


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failure than in LV tissue samples from unused donor hearts (*Figure 1A*). As shown by electrophoretic mobility shift assays, IRE binding activity was significantly reduced in failing hearts (most pronounced in patients with ischemic cardiomyopathy) (*Figure 1B*). Protein expression levels of the transferrin receptor were significantly lower in failing hearts than in the controls (*Figure 1C*).

Targeted Irp deletion in mice induces ID in the myocardium

We generated mice with a cardiomyocyte-targeted deletion of *Irp1* and *Irp2* (Cre-Irp1/2^{ex}) to address Irp function in the heart (*Figure 2A*). Cre-Irp1/2^{ex} mice were born at the expected Mendelian inheritance ratio and survived into adulthood. Reverse transcriptase polymerase chain reaction on LV myocardium and isolated cardiomyocytes demonstrated near-complete Cre-mediated deletion of Irp1 and Irp2 mRNAs in cardiomyocytes from Cre-Irp1/2^{ex} mice compared with littermates lacking the Cre transgene (Irp1/2^{ex}) (*Figure 2B*). Irp1 and Irp2 protein expression was markedly reduced in LV myocardium and barely detectable in isolated cardiomyocytes from Cre-Irp1/2^{ex} mice (*Figure 2C and*

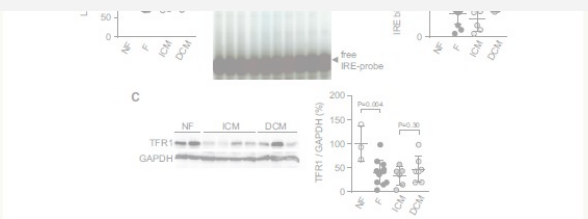
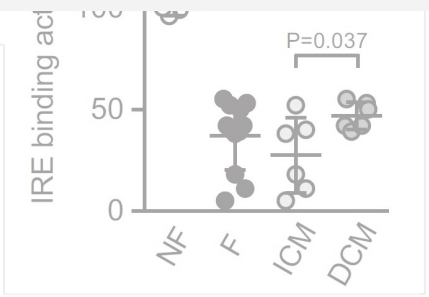


Figure 1 *Reduced IRP activity and iron content in failing human hearts.* (*A*) Non-haem iron concentration in left ventricular (LV) tissue samples from non-failing donors (NF) and patients with cardiac failure (F) due to ischemic cardiomyopathy (ICM) or dilated cardiomyopathy (DCM); $n = 4–6$ per group. (*B*) Representative electrophoretic mobility shift assay and summary data showing iron-responsive element (IRE) binding activity in LV tissue samples; $n = 4–6$ (control, no sample loaded). (*C*) Representative immunoblot and summary data showing transferrin receptor 1 (TFR1) and GAPDH protein expression in LV tissue samples; $n = 3–7$. *P* values were determined by two independent sample *t*-test.

Remember the paper with the t-test?



Iron-regulatory proteins secure iron availability in cardiomyocytes to prevent heart failure

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Statistically significant hypothesis test with IRE binding dependent variable on failing and non-failing hearts ($p<0.001$)

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40.0	96.0
38.0	100.0
18.0	105.0
11.0	
5.0	
42.0	
55.0	
53.0	
39.0	

« ‹ 1 of 2 › » Page 1 Show 10

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Data browser

Statistically significant hypothesis test with IRE binding dependent variable on failing and non-failing hearts (p<0.001)

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Iron-regulatory proteins secure iron availability in cardiomyocytes to prevent heart failure ★

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DOI: https://doi.org/10.1093/eurheartj/ehw333

Verified

Statistically significant hypothesis test wi...

Data browser

Statistically significant hypothesis test with IRE binding dependent variable on failing and non-failing (p<0.001)

Instance of: Contribution, obo:OBI_0000739 | Applied templates: Contribution, Student's t-test

has dependent variable	http://purl.obolibrary.org/obo/GO_0030350
has specified input	Summary data showing iron-responsive element (IRE) bi... tissue samples QBDataset
structure	Data Structure Definition C8
component	Component Specification failing heart (F) C6
measure	failing heart (F) C2, C5
order	2 Text
component	Component Specification non-failing heart (NF) C6
measure	non-failing heart (NF) C2, C5
order	1 Text
has specified output	the p-value of the statistical hypothesis test (p<0.001) C1003
value specification	0.0000000131112475 C1004
specified numeric value	0.0000000131112475 Text

iron-responsive element binding

Term Information

Accession GO:0030350 Data health

Name iron-responsive element binding

Ontology molecular_function

Synonyms IRE binding

Alternate IDs None

Definition Binding to an iron-responsive element, a regulatory sequence found in the 5'- and 3'-untranslated regions of mRNAs encoding many iron-binding proteins. *Source:* PMID:8710843, PMID:3198610

Comment None

History See term history for GO:0030350 at QuickGO

Subset None

Related
- Link to all **genes and gene products** annotated to iron-responsive element binding (**excluding "regulates"**).
- Link to all direct and indirect **annotations** to iron-responsive element binding (**excluding "regulates"**).
- Link to all direct and indirect **annotations download** (limited to first 10,000) for iron-responsive element binding (**excluding "regulates"**).

Include "regulates"

For more information, please see the ontology relation documentation.

Some remarks

- The ORKG content is generally extracted from published articles
- It is primarily manually created and curated by researchers
- This is typical for crowdsourced systems (e.g. Wikipedia, OpenStreetMaps)
- ORKG is thus an open and collaborative platform
- Additionally, the content can be extracted algorithmically from articles
- Traditionally with NLP-based approaches, now also LLM-based
- Likely most of the content is collected from already structured sources
- A major example of this is the data sourced from paperswithcode.com
- Uses “anyone can say anything about anything” (Semantic) Web principle
- Strong freedoms, but does it work for scientific knowledge?

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Give it a try at orkg.org

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The possibilities

Data browser

Contribution 1

Instance of: R40006, Contribution | Applied templates: Basic reproduction number estimate, Contribution

Referred: 3 times

Basic reproduction number	3.1 C5001
value	3.1 Text
Confidence interval (95%)	2.9 - 3.2 C2003
Lower confidence limit	2.9 Decimal
Upper confidence limit	3.2 Text
location	Lombardy, Italy DCLocation
same as	https://lobid.org/gnd/4036294-2 Text
same as	http://www.wikidata.org/entity/Q1210 Text
same as	https://sws.geonames.org/3174618/ Text
research problem	Determination of the COVID-19 basic reproduction number Problem
description	This research problem aims at determining the basic reproduction rate of COVID-19. Text
same as	https://en.wikipedia.org/wiki/COVID-19_pandemic Text
Time period	2020-01-14 - 2020-03-08 C2005
has beginning	2020-01-14 Text
has end	2020-03-08 Text

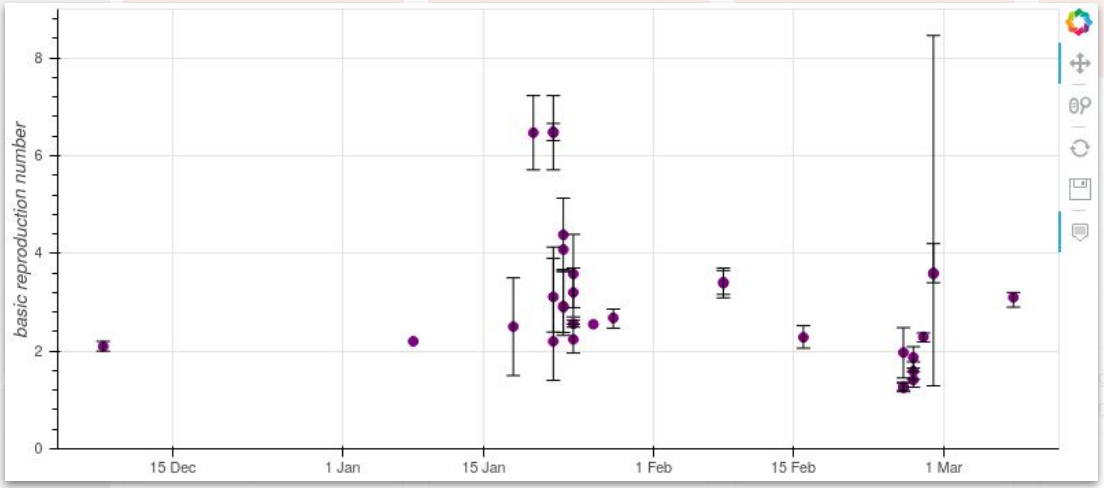
Given structured research findings published in individual papers …

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Properties	The early phase of the COVID-19 outbreak in Lombardy, Italy *2020 - Contribution 1*	Transmission potential of COVID-19 in Iran *2020 - Contribution 1*	Transmission potential of COVID-19 in Iran *2020 - Contribution 2*	Estimating the generation interval for COVID-19 based on symptom onset data *2020 - Contribution 1*
location	Lombardy, Italy	Iran	Iran	Singapore
Time period	Time interval	Time interval	Time interval	Time interval
has beginning	2020-01-14	2020-02-19	2020-02-19	2020-01-21
has end	2020-03-08	2020-02-29	2020-02-29	2020-02-26
Basic reproduction number	Basic reproduction number estimate value specification	Basic reproduction number estimate value specification	Basic reproduction number estimate value specification	Basic reproduction number estimate value specification
Has value	3.1	3.6	3.58	1.27
Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)
Lower confidence limit	2.9	3.4	1.29	1.19
Upper confidence limit	3.2	4.2	8.46	1.36

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Properties	The early phase of the COVID-19 outbreak in Lombardy, Italy *2020 - Contribution 1*	Transmission potential of COVID-19 in Iran *2020 - Contribution 1*	Transmission potential of COVID-19 in Iran *2020 - Contribution 2*	Estimating the generation interval for COVID-19 based on symptom onset data *2020 - Contribution 1*
location				Singapore
Time period				Time interval
has beginning				2020-01-21
has end				2020-02-26
Basic reproduction nu				t reproduction number nate value specification
Has value				1.27
Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)
Lower confidence limit	2.9	3.4	1.29	1.19
Upper confidence limit	3.2	4.2	8.46	1.36



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```
[ ]: import requests
     import datetime
     import pandas as pd
     import numpy as np
     from orkg import ORKG
     from bokeh.io import export_png
     from bokeh.models import ColumnDataSource, HoverTool, WheelZoomTool, ResetTool, SaveTool, PanTool, DatetimeTickFormatter, Whisker
     from bokeh.plotting import figure, show, output_notebook

     output_notebook()
```

```
[ ]: orkg = ORKG(host='https://orkg.org/orkg', simcomp_host='https://orkg.org/orkg/simcomp')

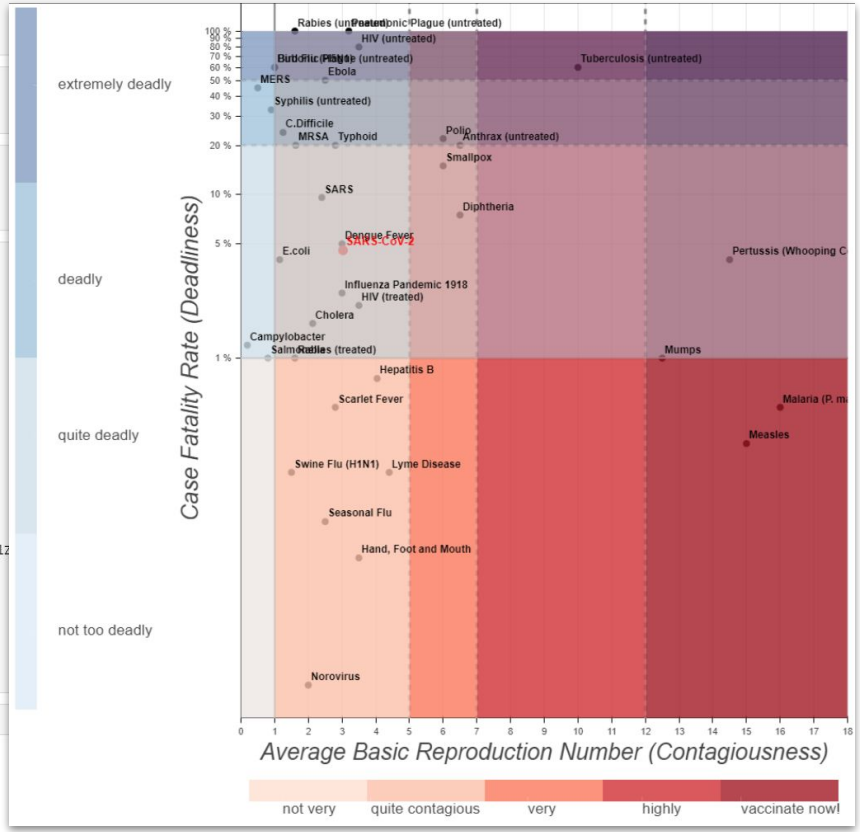
     df = orkg.contributions.compare_dataframe(comparison_id='R44930')
```

```
[ ]: dates = np.array([datetime.date.fromisoformat(x) for x in df.loc['has end', :]])
     values = np.float32(df.loc['Has value', :])
     lower = np.array([np.float32(x) if x else np.nan for x in df.loc['Lower confidence limit', :]])
     upper = np.array([np.float32(x) if x else np.nan for x in df.loc['Upper confidence limit', :]])
```

```
[ ]: hover1 = HoverTool(
         tooltips=[
             ('Date', '@date{%F}'),
             ('R0', '@value{0.ff}'),
             ('95% CI', '@lower{0.ff}-@upper{0.ff}')
         ],
         formatters={
             '@date': 'datetime',
             '@{value}' : 'printf',
             '@{lower}' : 'printf',
             '@{upper}' : 'printf'
         }
     )

     df = pd.DataFrame(data=dict(date=dates, value=values, lower=lower, upper=upper))
     source = ColumnDataSource(df)
     p = figure(x_axis_type="datetime", y_range=(0, 9), plot_width=800, plot_height=350, tools=[hover1, WheelZ
     p.xaxis.formatter=DatetimeTickFormatter(days=['%d %b'])
     p.yaxis.axis_label = 'basic reproduction number'
     p.circle('date', 'value', source=source, size=7, color='purple')
     p.add_layout(
         Whisker(source=source, base='date', upper='upper', lower='lower', level='overlay')
     )
     show(p)
```

```
[ ]: export_png(p, filename='img/R0-estimates-plot.png')
```



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Benchmark | Object detection on Coco test-dev

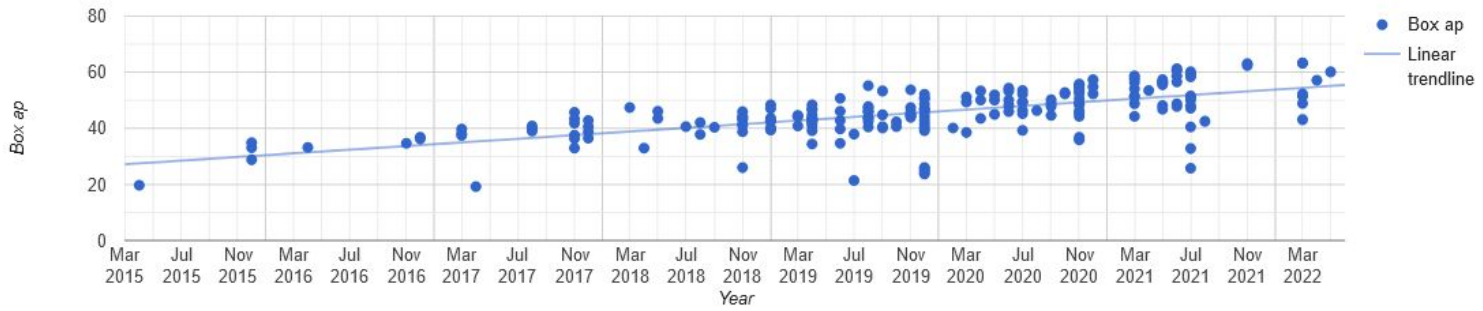
Edit

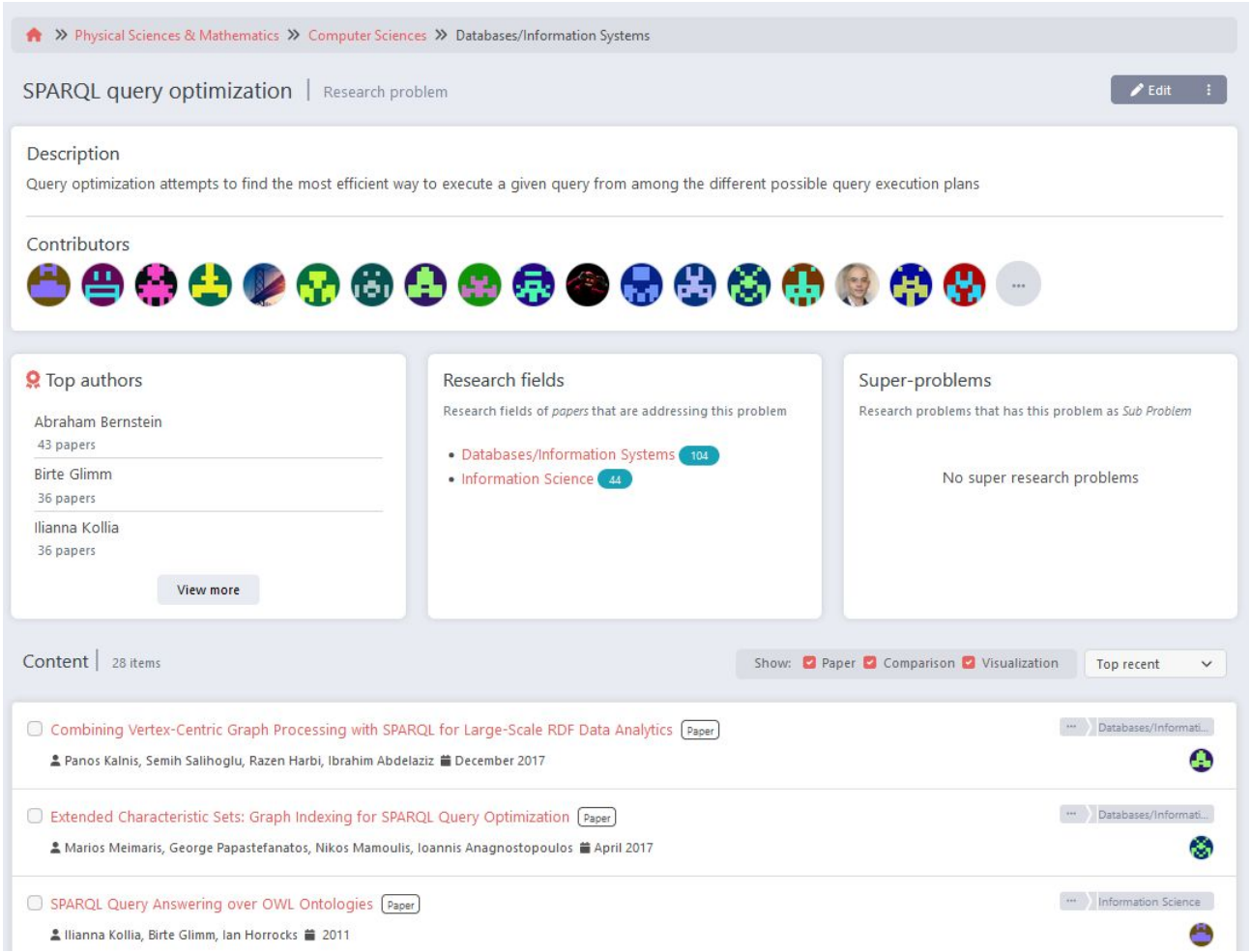
Research problem: Object detection

Dataset: Coco test-dev

Performance trend

Research problem: Object detection | Metric: Box ap




Physical Sciences & Mathematics » Computer Sciences » Databases/Information Systems
SPARQL query optimization | Research problem
Edit
Description
Query optimization attempts to find the most efficient way to execute a given query from among the different possible query execution plans
Contributors
Top authors
Abraham Bernstein
43 papers
Birte Glimm
36 papers
Ilianna Kollia
36 papers
View more
Research fields
Research fields of papers that are addressing this problem
Databases/Information Systems 104
Information Science 44
Super-problems
Research problems that has this problem as Sub Problem
No super research problems
Content | 28 items
Show: Paper Comparison Visualization
Top recent
Combining Vertex-Centric Graph Processing with SPARQL for Large-Scale RDF Data Analytics Paper
Panos Kalnis, Semih Salihoglu, Razen Harbi, Ibrahim Abdelaziz December 2017
Databases/Informati...
Extended Characteristic Sets: Graph Indexing for SPARQL Query Optimization Paper
Marios Meimaris, George Papastefanatos, Nikos Mamoulis, Ioannis Anagnostopoulos April 2017
Databases/Informati...
SPARQL Query Answering over OWL Ontologies Paper
Ilianna Kollia, Birte Glimm, Ian Horrocks 2011
Information Science

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Data access

REST API

You can access the Data by using the our REST API.

API Documentation

Python Package

Python package wrapping the ORKG API

Python Package

SPARQL endpoint

You can access the following SPARQL endpoint to explore our data using SPARQL queries.

Virtuoso Query Editor | Visual SPARQL Editor

Export data: RDF Dump

You can download the Open Research Knowledge Graph RDF Dump in N-Triples format below

Download RDF dump

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Vergangene Woche lag der Anteil noch bei 55 Prozent, zu Beginn der RKI-Erhebung vor rund anderthalb Monaten lediglich bei sechs Prozent. Überraschend ist der schnelle Anstieg nicht, denn schon von der ersten Entdeckung an galt die britische Variante als deutlich ansteckender als der zuvor vorherrschende Typ des Virus. Zunächst ging man davon aus, dass ein Infizierter im Schnitt - je nach Untersuchung - 50 bis 70 Prozent mehr Menschen ansteckt, später wurde dieser Wert auf etwa 35 Prozent korrigiert. Das mag auf den ersten Blick nicht sonderlich dramatisch klingen, doch aufgrund des exponentiellen Wachstums droht in kurzer Zeit eine neue Dimension der Ausbreitung.

Variante ist bis zu 67 Prozent häufiger tödlich

Wesentlich bedeutsamer sind jedoch die neuesten Erkenntnisse, was die Gefährlichkeit von B.1.1.7 angeht. Anfangs hieß es, die Mutante sei nicht tödlicher als die bisherigen Coronavirus-Formen. Doch mittlerweile sind in Großbritannien drei Studien erschienen (ein Preprint aus London , eine Studie aus Exeter und eine auf "Nature" veröffentlichte und bereits begutachtete Studie), die ein alarmierendes Bild zeichnen. Demnach ist die britische Variante nicht nur ansteckender, sondern auch deutlich häufiger tödlich. Das Risiko, innerhalb von 28 Tagen nach der Infektion zu versterben, ist - je nach Studie - um 58 bis 67 Prozent erhöht gegenüber einem Patienten, der eine der bisherigen Corona-Varianten hat.

https://www.tagesschau.de/faktenfinder/britische-mutante-b117-101.html
(Available via Internet Archive)

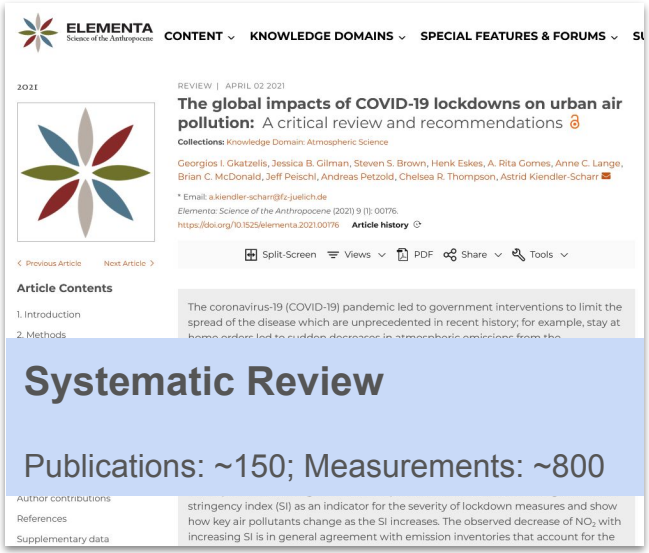
Vergangene Woche lag der Anteil noch bei 55 Prozent, zu Beginn der RKI-Erhebung vor rund anderthalb Monaten lediglich bei sechs Prozent. Überraschend ist der schnelle Anstieg nicht, denn schon von der ersten Entdeckung an galt die britische Variante als deutlich ansteckender als der zuvor vorherrschende Typ des Virus. Zunächst ging man davon aus, dass ein Infizierter im Schnitt - je nach Untersuchung - 50 bis 70 Prozent mehr

Mensch
Das ma
aufgrun
Dimens

Varian

Wesent
Gefähr
tödlich
Großbr
Studie
beguta
die brit
tödlich
ist - je
der ein

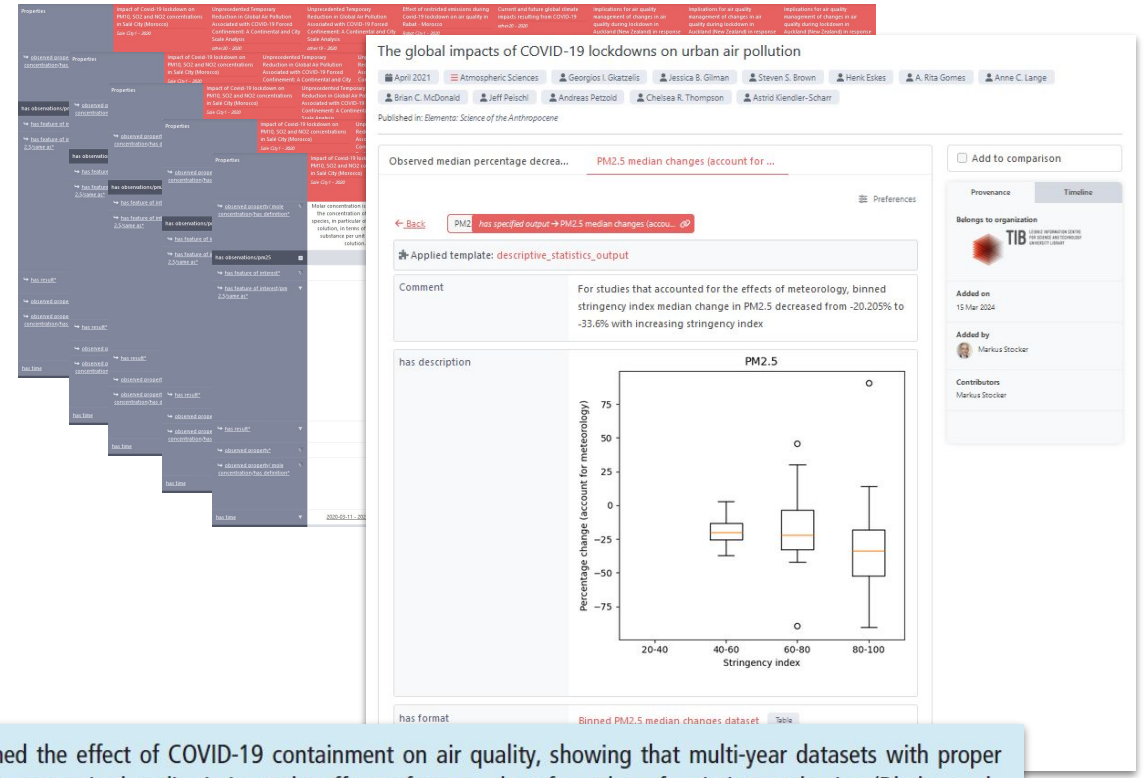
Properties	Case fatality risk of the SARS-CoV-2 variant of concern B.1.1.7 in England *Contribution 1 - 2021*	Increased mortality in community-tested cases of SARS-CoV-2 lineage B.1.1.7 *Contribution 1 - 2021*	Risk of mortality in patients infected with SARS-CoV-2 variant of concern 202012/1: matched cohort study *Contribution 1*
is about	SARS-CoV-2 variant of concern B.1.1.7	SARS-CoV-2 variant of concern B.1.1.7	SARS-CoV-2 variant of concern B.1.1.7
hazard ratio	Case fatality rate estimate value specification	Case fatality rate estimate value specification	Case fatality rate estimate value specification
Has value	1.67	1.55	1.64
Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)	Confidence interval (95%)
Lower confidence limit	1.34	1.39	1.32
Upper confidence limit	2.09	1.72	2.04
location	England	England	United Kingdom
Time period	Time interval	Time interval	Time interval
has beginning	2020-11-16	2020-09-01	2020-10-01
has end	2021-01-11	2021-02-14	2021-02-12



Systematic Review

Publications: ~150; Measurements: ~800

https://doi.org/10.1525/elementa.2021.00176



Several studies have examined the effect of COVID-19 containment on air quality, showing that multi-year datasets with proper statistical/modelling analysis are required to discriminate the effects of meteorology from that of emissions reduction (Dhaka et al., 2020; L. Li et al., 2020; Wang et al., 2020; Zhao et al., 2020b). Accounting for meteorological influences and with an increasing stringency index, the median observed change in NO_2 decreased from –13% to –48%, and in $PM_{2.5}$ decreased from –10% to –33%, whereas the median change in ozone increased from 0% to 4% (Gkatzelis et al., 2021). The latter can be explained by the decrease of NO emissions that titrate ozone in specific highly polluted areas, leading to the observed increase in surface ozone concentration in cities (Le et al., 2020; Sicard et al., 2020; Huang et al., 2021).

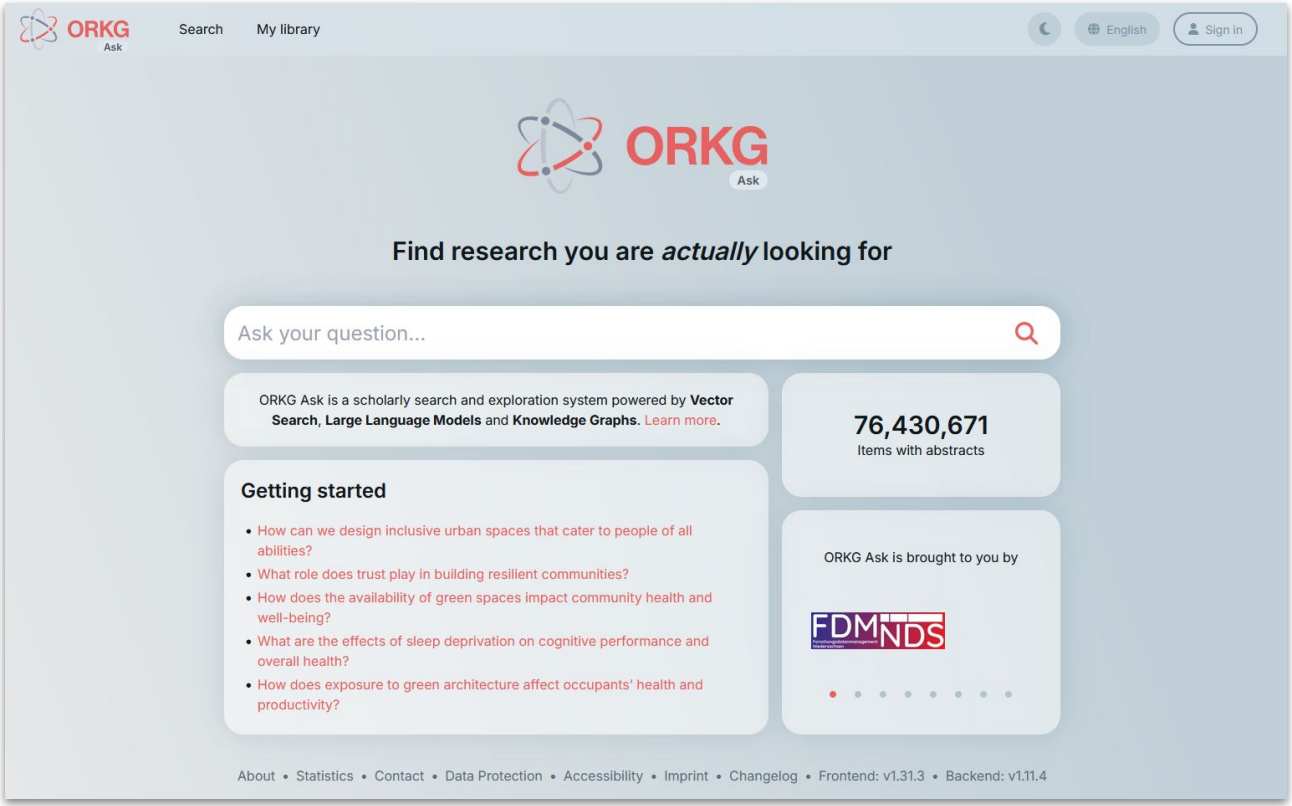
ipcc

https://www.ipcc.ch/report/ar6/wg1/chapter/chapter-6/#cross-chapter-box-6.1

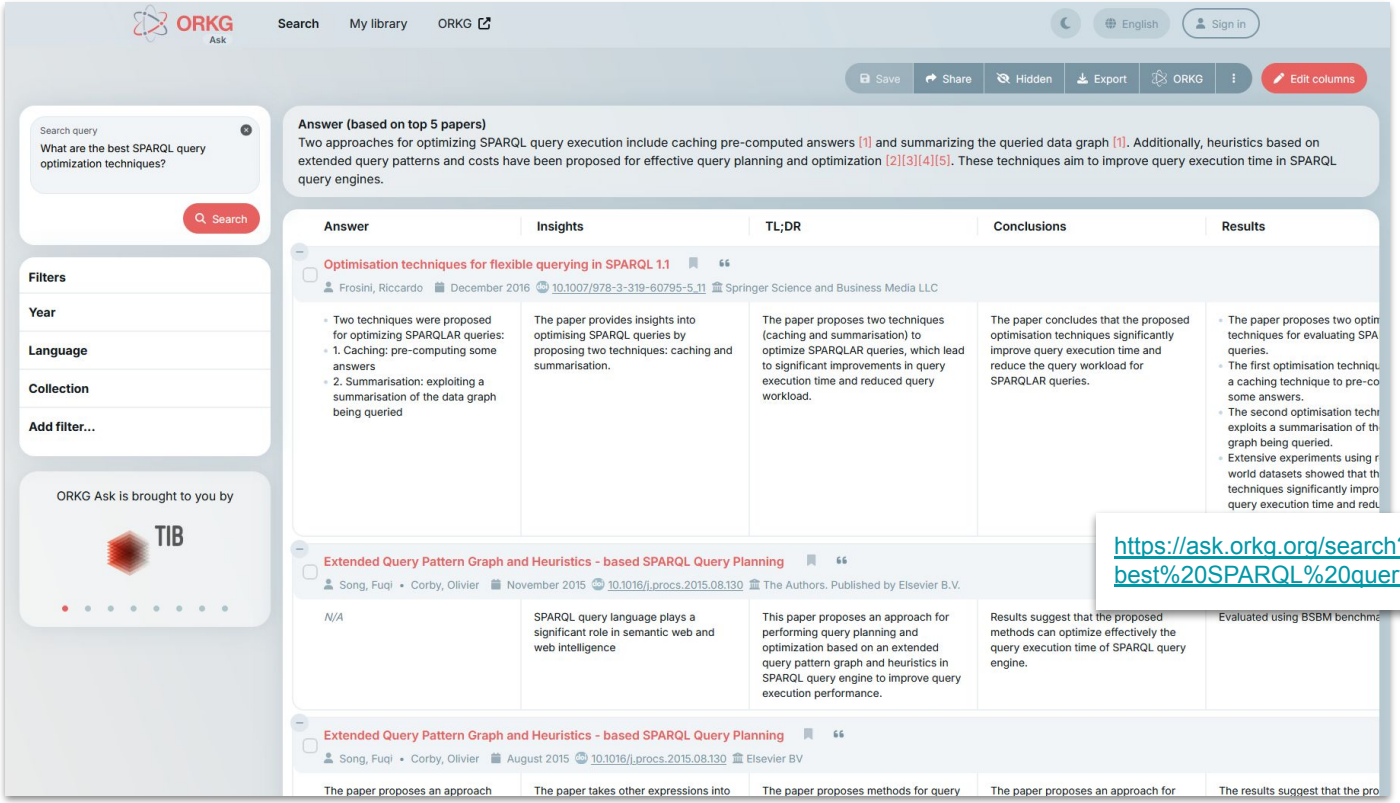
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Possibilities with GenAI

ORKG Ask

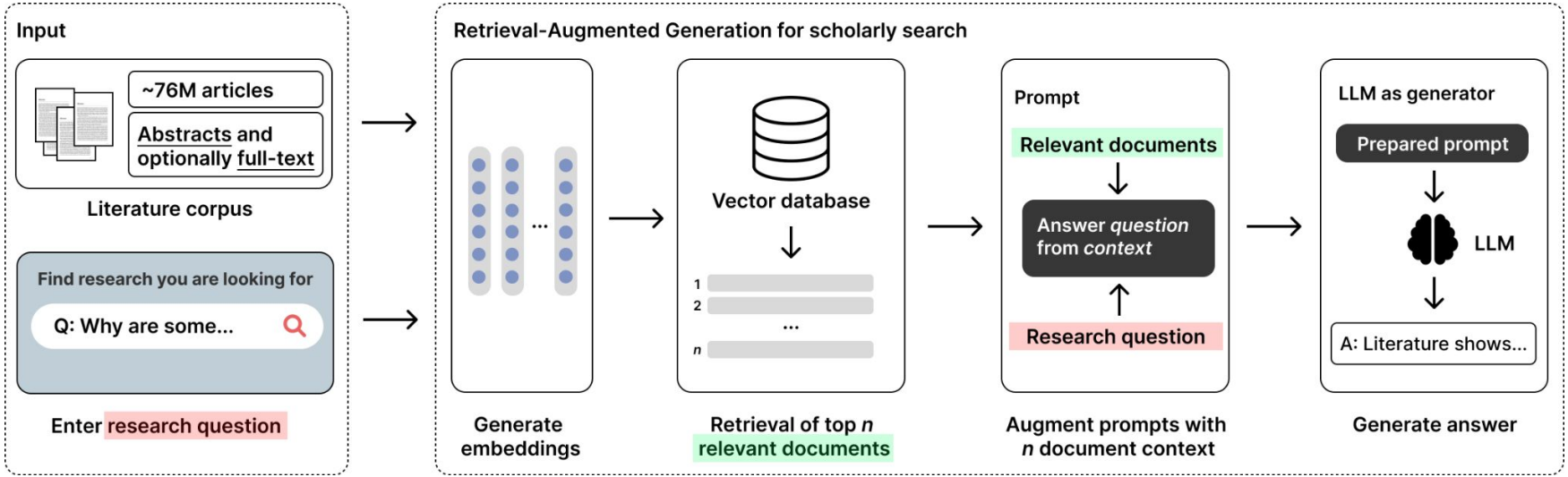

ORKG
Ask
Search
My library
English
Sign in
ORKG
Ask
Find research you are *actually* looking for
Ask your question...
ORKG Ask is a scholarly search and exploration system powered by **Vector Search**, **Large Language Models** and **Knowledge Graphs**. Learn more.
76,430,671
Items with abstracts
Getting started
How can we design inclusive urban spaces that cater to people of all abilities?
What role does trust play in building resilient communities?
How does the availability of green spaces impact community health and well-being?
What are the effects of sleep deprivation on cognitive performance and overall health?
How does exposure to green architecture affect occupants' health and productivity?
ORKG Ask is brought to you by
FDM NDS
About • Statistics • Contact • Data Protection • Accessibility • Imprint • Changelog • Frontend: v1.31.3 • Backend: v1.11.4

ORKG Ask



https://ask.orkg.org/search?query=What%20are%20the%20best%20SPARQL%20query%20optimization%20techniques

ORKG Ask

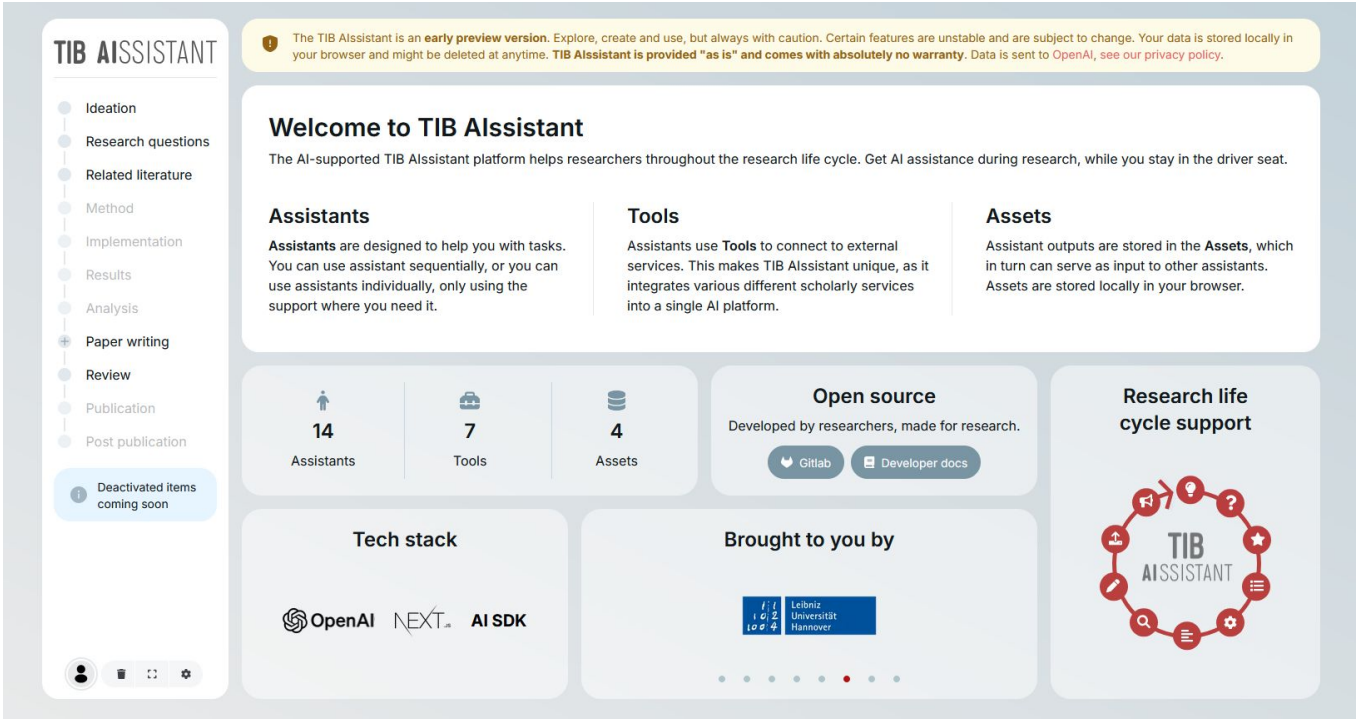


Oelen, A., Jaradeh, M. Y., & Auer, S. (2024). ORKG ASK: a Neuro-symbolic Scholarly Search and Exploration System. SEMANTiCS 2024.
https://ceur-ws.org/Vol-3759/

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Give it a try at ask.orkg.org

TIB AIssistant



Auer, S., et al. (2025). Towards AI-Supported Research: a Vision of the TIB AIssistant. Sci-K 2025. https://ceur-ws.org/Vol-4065/

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Questions

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Challenges

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Time-series Land Cover Mapping And Urban Expansion Analysis Using OpenStreetMap Data And Remote Sensing Big Data: A Case Study Of Guangdong-Hong Kong-Macao Greater Bay Area, China

2022 | 1 citation | Qing Ding | Zhenfeng Shao | Xiao Huang | Orhan Altan | Bin Hu

DOI: https://doi.org/10.1016/j.jag.2022.103001

NFDI4Ing Contribution

Preferences

Applied template: Contribution

cites repository	emmajschoenmakers/comparative_impact_irp-22

Add to comparison

Provenance | Timeline

Added on
05 Sep 2023

Added by
Vasiliy Seibert

Contributors
Vasiliy Seibert

Poor data quality



Note: The analysis is somewhat dated but the overall message still holds.

Data accuracy and reproducibility

Leibniz Universität Hannover

Statistically significant hypothesis test wi...

Data browser

Statistically significant hypothesis test with IRE binding dependent variable on failing and non-failing hearts (p<0.001)

Instance of: Contribution, obo:OBI_0000739 | Applied templates: Contribution, Student's t-test

has dependent variable	http://purl.obolibrary.org/obo/GO_0030350 Text
has specified input	Summary data showing iron-responsive ele... tissue samples QBDataset
structure	Data Structure Definition C8
component	Component Specification failing heart (F) C6
measure	failing heart (F) C2, C5
order	2 Text
component	Component Specification non-failing
measure	non-failing heart (NF) C2, C5
order	1 Text
has specified output	the p-value of the statistical
value specification	0.0000000131112475 C1004
specified numeric value	0.0000000131112475 Text

failing heart (F)	non-failing heart (NF)
52.0	99.0
40.0	96.0
38.0	100.0
18.0	105.0
11.0	
5.0	
42.0	
55.0	
53.0	
39.0	

« ‹ 1 of 2 › » Page 1 Show 10

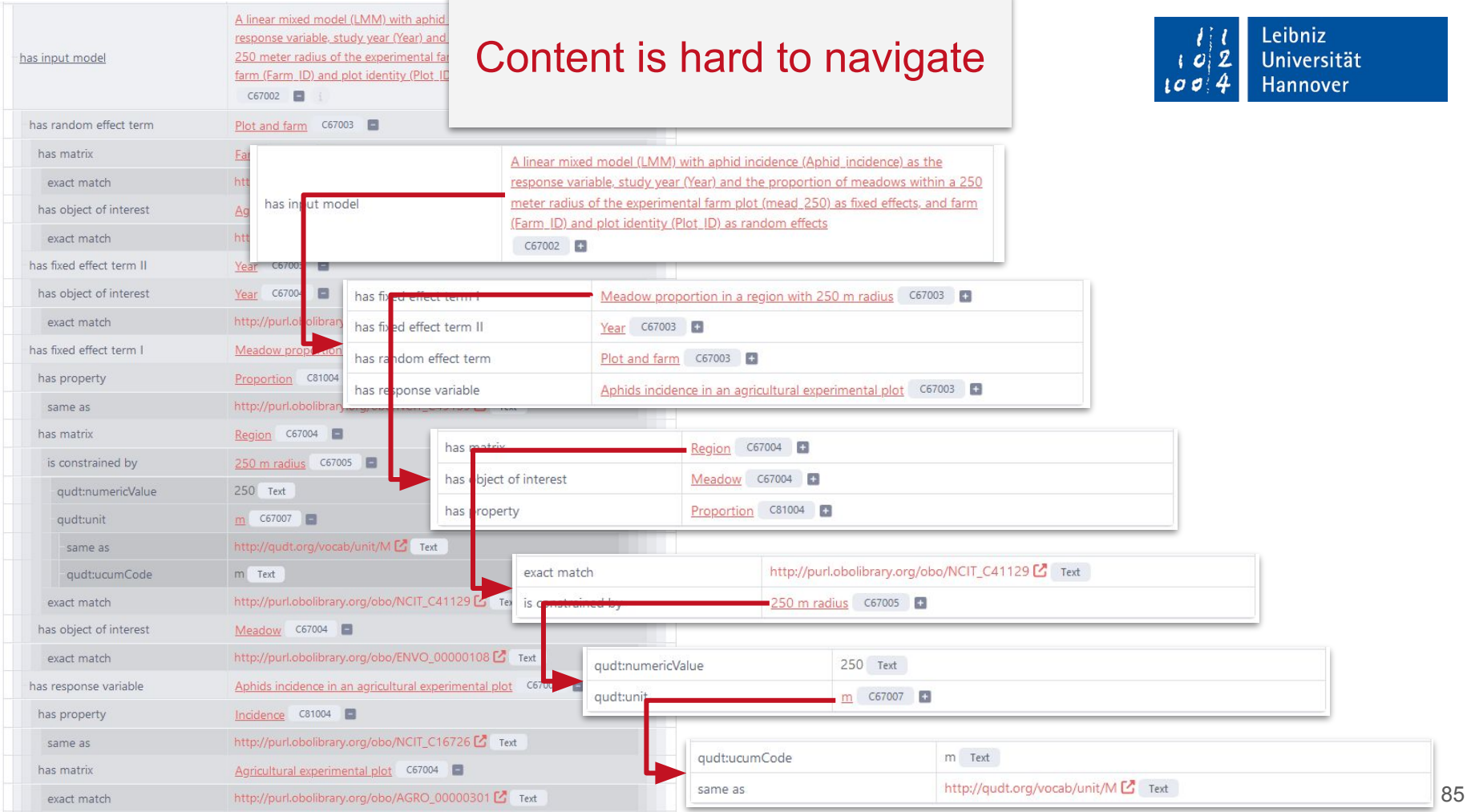
Stat has value specification → 0.0000000131112475

0.0000000131112475

Instance of: obo:OBI_0001931 | Applied template: Scalar value specification

specified numeric value	0.0000000131112475 Text

Content is hard to navigate



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Questions*

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TIB Knowledge Loom

Recap

- The ORKG content is extracted from published articles
- Either manually by experts or semi-automatically with machines
- Either way, it is expensive (hard to scale) and/or inaccurate
- Researchers are accurate, but hard to motivate and scale
- Plus, entering hierarchical data in a UI manually is a nightmare
- Users stumble before they even start: “How should I structure my data?”
- Algorithmic extraction could scale, but its accuracy is not great
- Plus, traditional NLP with supervised models is expensive and task specific
- Newer LLM-based approaches made some progress, but the problem remains
- Plus, all the prompt-engineering and evaluation is anything but automated
- Hence, the scalability remains untested in practice

You may ask …

- … wait, why don’t we produce structured knowledge from the outset?
- First we bury knowledge into narrative text documents
- Then we develop an expensive machinery to extract and structure it (again)
- Even though knowledge is (more) structured before we bury it
- This is the intuition underlying the TIB Knowledge Loom

What are Loom Articles

- Loom Records where scientific findings are born-reusable from the outset
- Research articles with a specialized data set as interlinked material
- The specialized data set consists of
  - A (RO-Crate) metadata file in JSON-LD format (ro-crate-metadata.json)
  - One or more statement data files in JSON-LD format
  - Additional files, in particular CSV data, images, and code in Python or R languages
- In the TIB Knowledge Loom, this specialized data set is a Loom Record

Excursion: RO-Crate

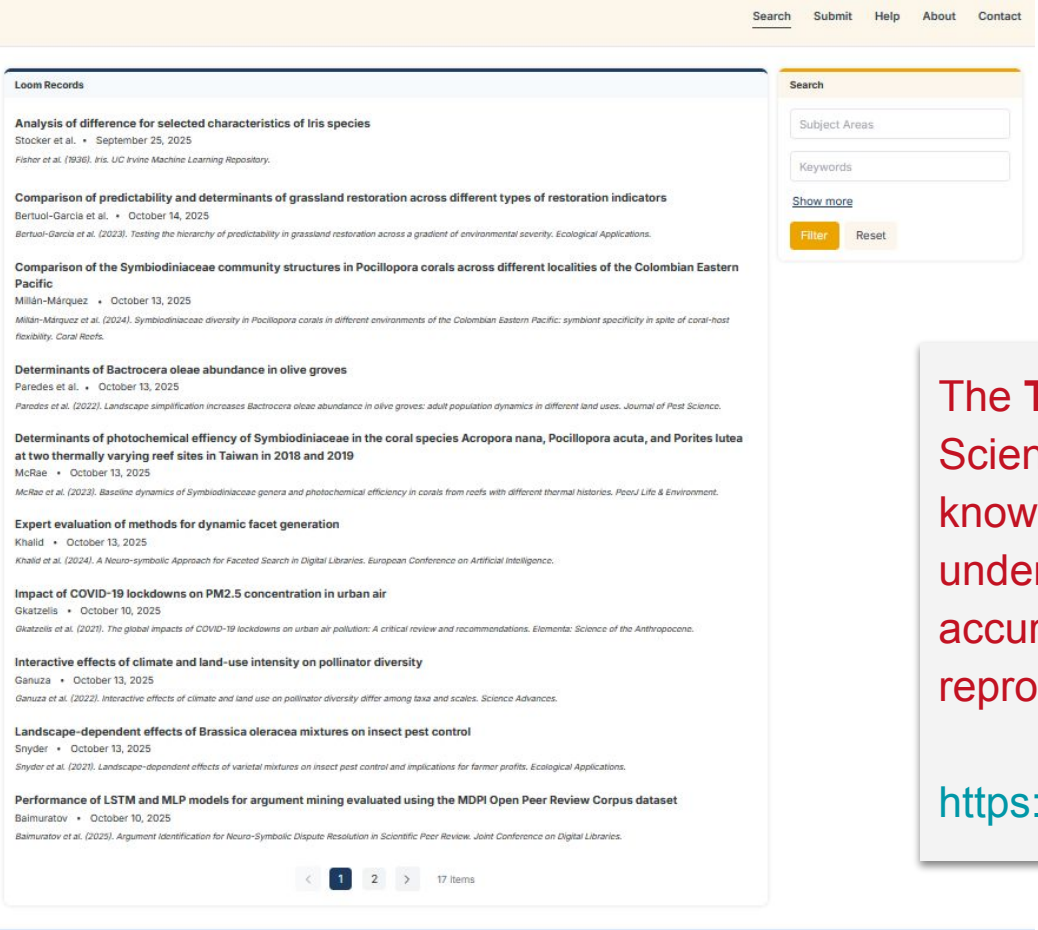
- A lightweight approach to packaging research data with their metadata
- Aims to help people make their research FAIR
- It describes the methods, data, output and outcomes of a piece of work
- Linking this together enables sharing research outputs with their context
- https://www.researchobject.org/ro-crate/

Excursion: JSON-LD

- JSON is the JavaScript Object Notation (see https://www.json.org)
- A lightweight data-interchange format
- JSON-LD is a lightweight Linked Data format (see https://json-ld.org)
- A JSON-LD document is a serialization of an RDF dataset
- RDF is the Resource Description Framework (see https://www.w3.org/RDF)
- RDF is a graph-based data model organizing data in Statements
- Statement is a triple consisting of a subject, a predicate, and an object
- Example (readable and formal)
  - (Leibniz University Hannover, location, Hannover)
  - (Hannover, country, Germany)
  - (wikidata:Q678982, wikidata:P276, wikidata:Q1715)
  - (wikidata:Q1715, wikidata:P17, wikidata:Q183)

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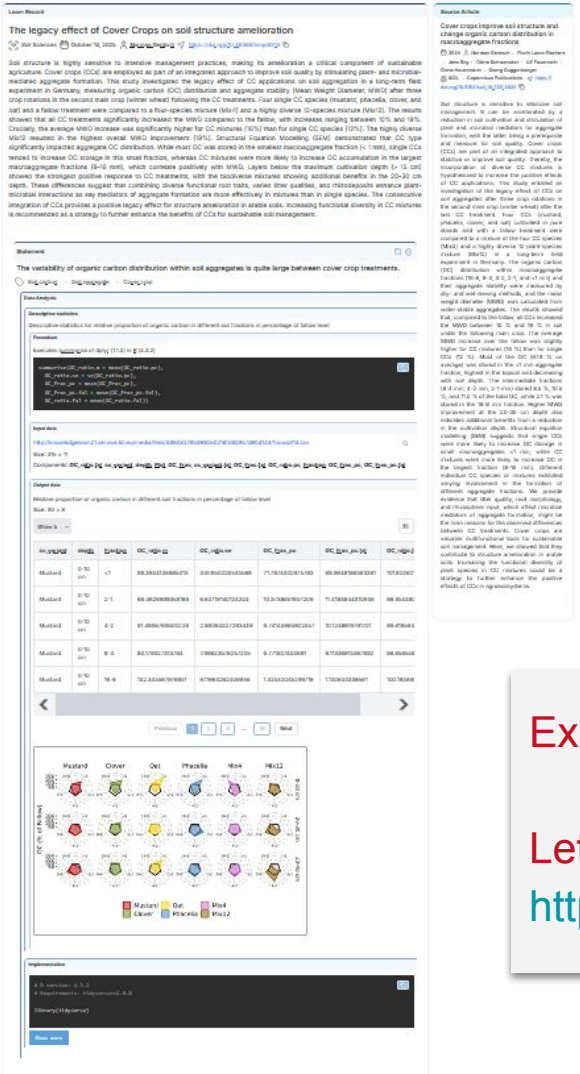
Give it a try wikidata.org/entity/Q678982



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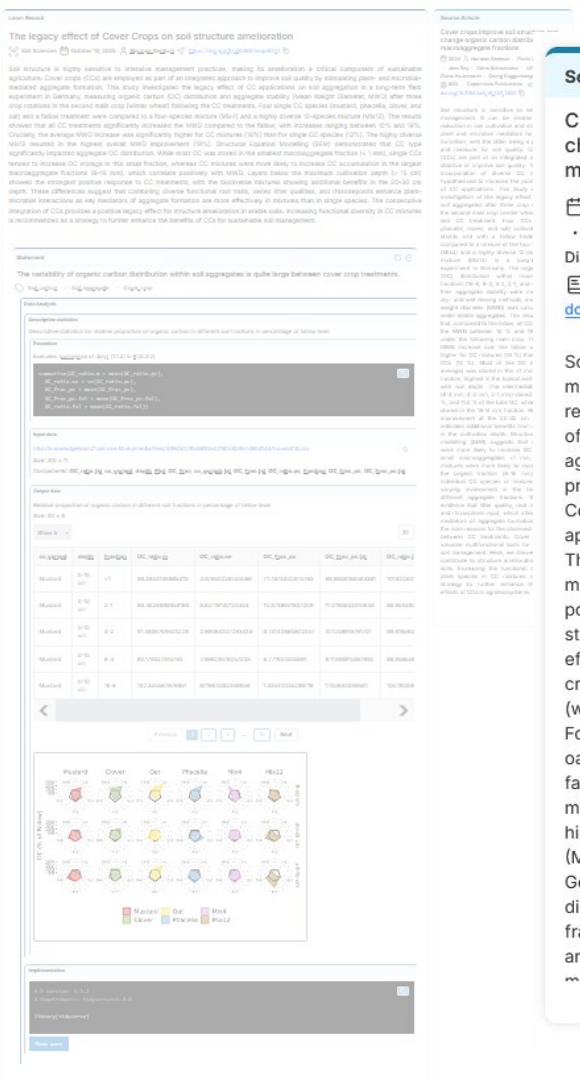
The **TIB Knowledge Loom** is an Open Science Digital Library for AI-ready scientific knowledge where scientific findings and underlying evidence are presented clearly, accurately, in rich structured format, and in reproducible manner.

https://knowledgeloom.tib.eu



Example Loom Record

Let's look at the original first
https://doi.org/10.5194/soil-10-139-2024



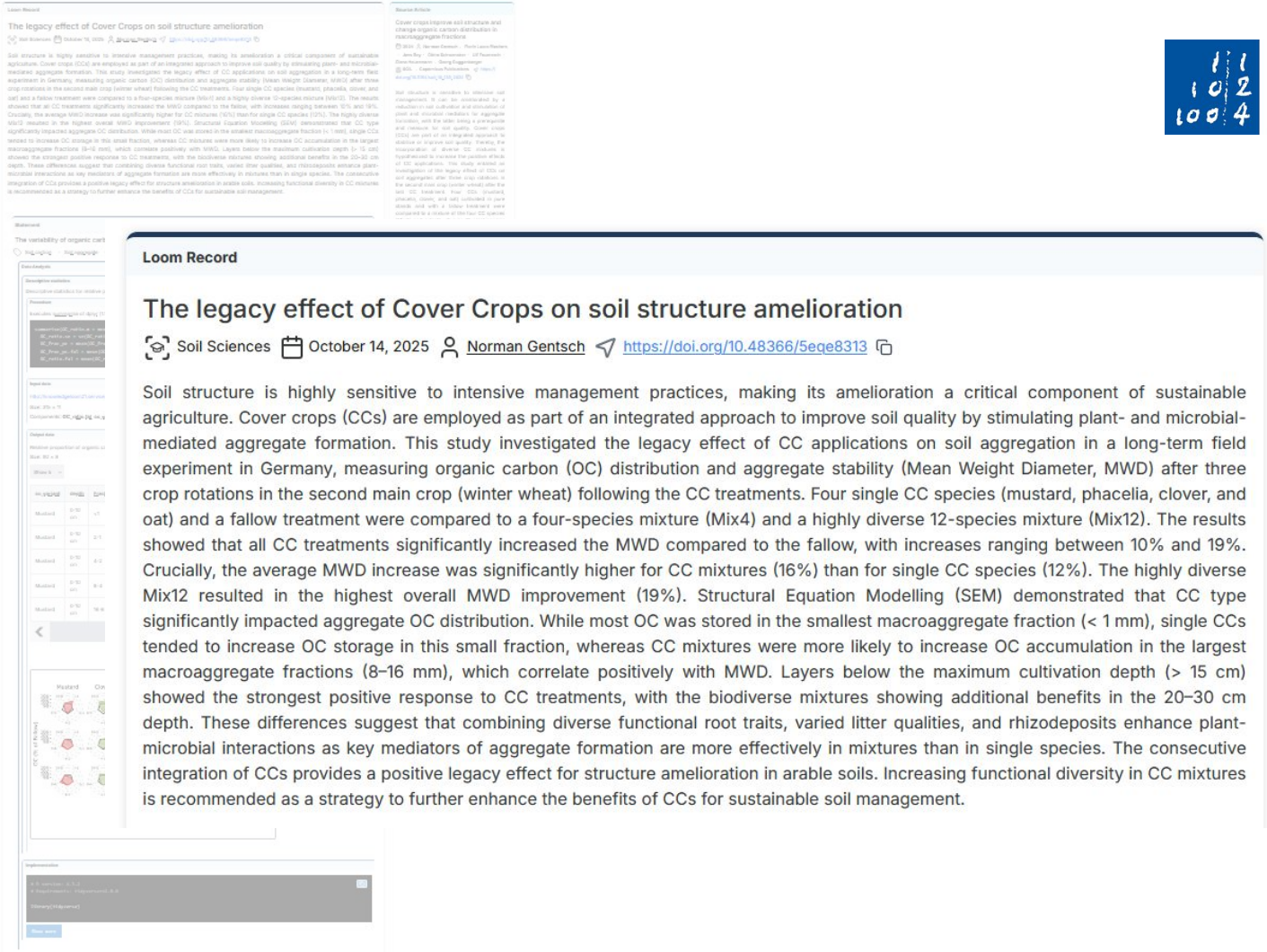
Source Article

Cover crops improve soil structure and change organic carbon distribution in macroaggregate fractions

2024 Norman Gentsch · Florin Laura Riechers · Jens Boy · Dörte Schweneker · Ulf Feuerstein · Diana Heuermann · Georg Guggenberger

SOIL · Copernicus Publications https://doi.org/10.5194/soil-10-139-2024

Soil structure is sensitive to intensive soil management. It can be ameliorated by a reduction in soil cultivation and stimulation of plant and microbial mediators for aggregate formation, with the latter being a prerequisite and measure for soil quality. Cover crops (CCs) are part of an integrated approach to stabilize or improve soil quality. Thereby, the incorporation of diverse CC mixtures is hypothesized to increase the positive effects of CC applications. This study entailed an investigation of the legacy effect of CCs on soil aggregates after three crop rotations in the second main crop (winter wheat) after the last CC treatment. Four CCs (mustard, phacelia, clover, and oat) cultivated in pure stands and with a fallow treatment were compared to a mixture of the four CC species (Mix4) and a highly diverse 12-plant-species mixture (Mix12) in a long-term field experiment in Germany. The organic carbon (OC) distribution within macroaggregate fractions (16-8, 8-4, 4-2, 2-1, and <1 mm) and their aggregate stability were measured by dry and wet sieving



Loom Record

The legacy effect of Cover Crops on soil structure amelioration

Soil Sciences October 14, 2025 Norman Gentsch https://doi.org/10.48366/5eqe8313

Soil structure is highly sensitive to intensive management practices, making its amelioration a critical component of sustainable agriculture. Cover crops (CCs) are employed as part of an integrated approach to improve soil quality by stimulating plant- and microbial-mediated aggregate formation. This study investigated the legacy effect of CC applications on soil aggregation in a long-term field experiment in Germany, measuring organic carbon (OC) distribution and aggregate stability (Mean Weight Diameter, MWD) after three crop rotations in the second main crop (winter wheat) following the CC treatments. Four single CC species (mustard, phacelia, clover, and oat) and a fallow treatment were compared to a four-species mixture (Mix4) and a highly diverse 12-species mixture (Mix12). The results showed that all CC treatments significantly increased the MWD compared to the fallow, with increases ranging between 10% and 19%. Crucially, the average MWD increase was significantly higher for CC mixtures (16%) than for single CC species (12%). The highly diverse Mix12 resulted in the highest overall MWD improvement (19%). Structural Equation Modelling (SEM) demonstrated that CC type significantly impacted aggregate OC distribution. While most OC was stored in the smallest macroaggregate fraction (< 1 mm), single CCs tended to increase OC storage in this small fraction, whereas CC mixtures were more likely to increase OC accumulation in the largest macroaggregate fractions (8–16 mm), which correlate positively with MWD. Layers below the maximum cultivation depth (> 15 cm) showed the strongest positive response to CC treatments, with the biodiverse mixtures showing additional benefits in the 20–30 cm depth. These differences suggest that combining diverse functional root traits, varied litter qualities, and rhizodeposits enhance plant-microbial interactions as key mediators of aggregate formation are more effectively in mixtures than in single species. The consecutive integration of CCs provides a positive legacy effect for structure amelioration in arable soils. Increasing functional diversity in CC mixtures is recommended as a strategy to further enhance the benefits of CCs for sustainable soil management.


The legacy effect of Cover Crops on soil structure amelioration

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Statement

The variability of organic carbon distribution within soil aggregates is quite large between cover crop treatments.

Soil carbon • Soil aggregate • Cover crop

Statement

When focusing on individual soil horizons, pairwise comparison indicates significantly higher mean weight diameter (MWD) compared to the fallow treatment for clover at 0-10 cm (18.8% higher, p=0.017), Mix12 at 20-30 cm (37.6% higher, p=0.049), and phacelia (17.0% higher, p=0.018) and Mix4 (12.8% higher, p=0.037) at 30-40 cm.

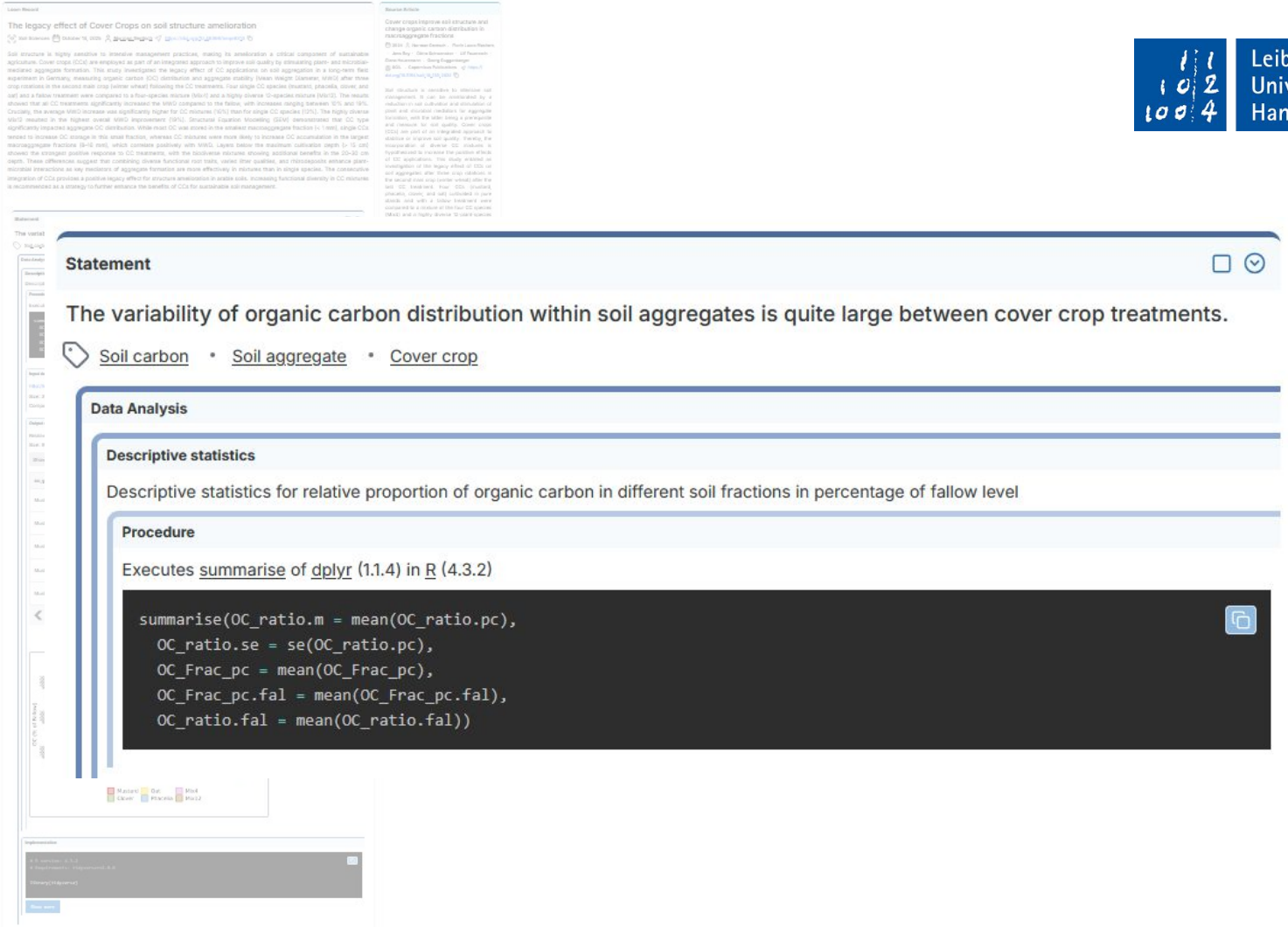
Soil horizon • Mean weight diameter (MWD) of soil aggregates • Fallow • Phacelia

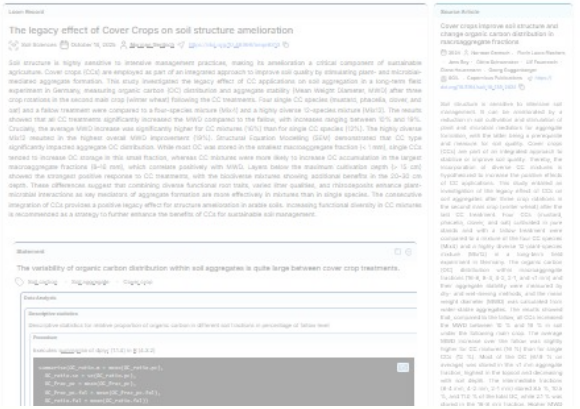
Statement

Mean weight diameter (MWD) of soil aggregates increases with soil depth and is significantly higher in the cover crop (CC) treatments than in the fallow treatment.

Soil aggregate • Cover crop • Mean weight diameter (MWD) of soil aggregates • Fallow




Statement
The variability of organic carbon distribution within soil aggregates is quite large between cover crop treatments.
Soil carbon • Soil aggregate • Cover crop
Data Analysis
Descriptive statistics
Descriptive statistics for relative proportion of organic carbon in different soil fractions in percentage of fallow level
Procedure
Executes summarise of dplyr (1.1.4) in R (4.3.2)
summarise(OC_ratio.m = mean(OC_ratio.pc),
OC_ratio.se = se(OC_ratio.pc),
OC_Frac_pc = mean(OC_Frac_pc),
OC_Frac_pc.fal = mean(OC_Frac_pc.fal),
OC_ratio.fal = mean(OC_ratio.fal))

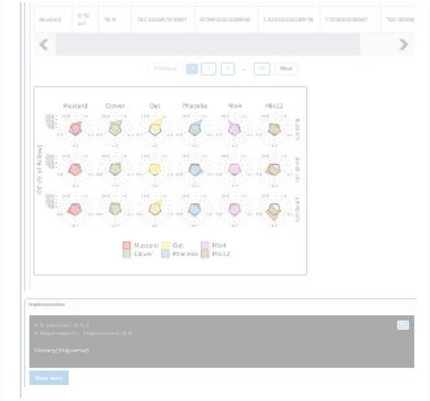


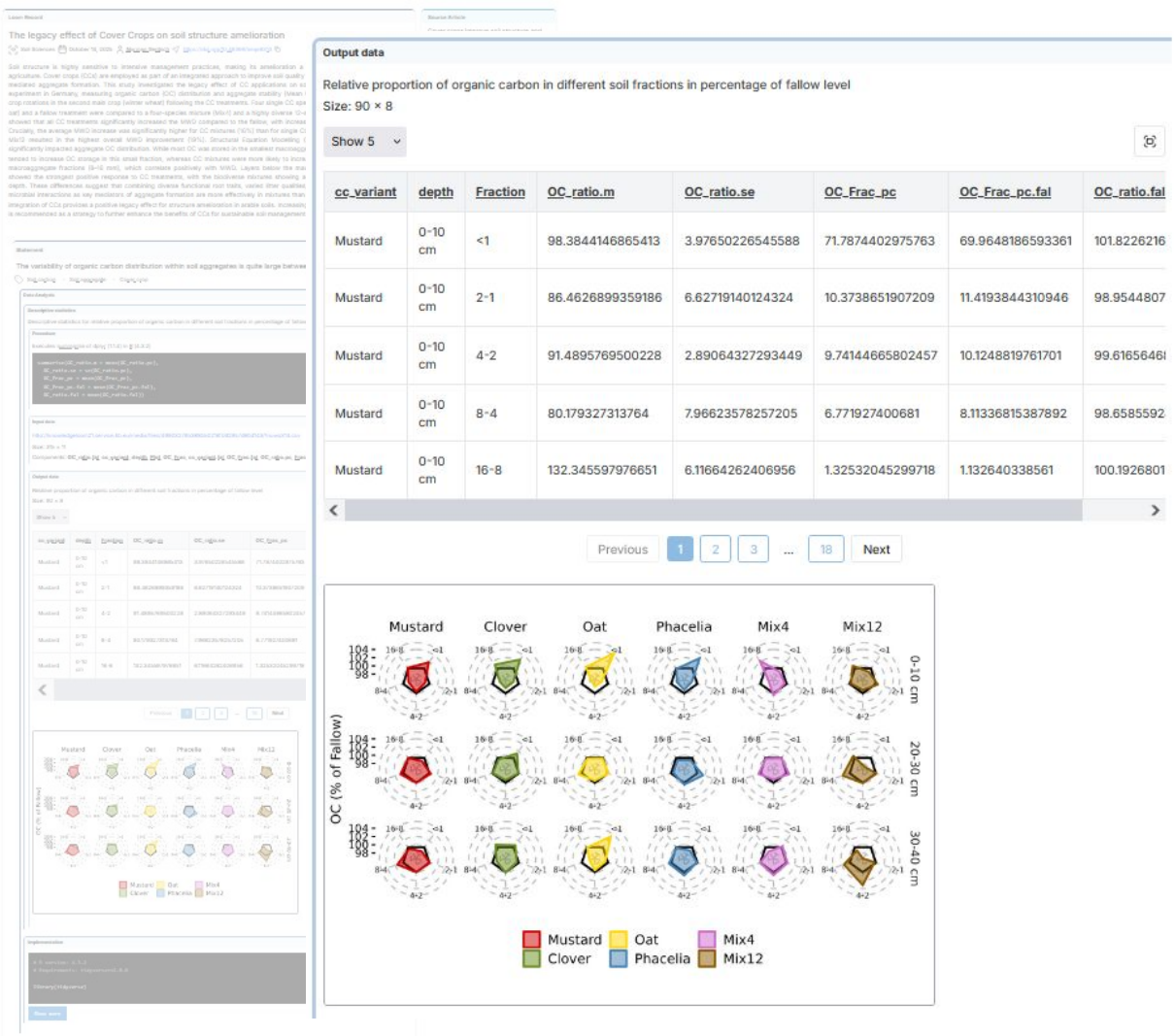
Input data

http://knowledgeloom21.service.tib.eu/media/files/49903279b388b50216fb92957d60d143/1nuwo314.csv

Size: 315 × 11

Components: **OC_ratio.fal**, **cc_variant**, **depth**, **Plot**, **OC_Frac**, **cc_variant.fal**, **OC_Frac.fal**, **OC_ratio.pc**, **Fraction**, **OC_Frac_pc**, **OC_Frac_pc.fal**




The legacy effect of Cover Crops on soil structure amelioration
Output data
Relative proportion of organic carbon in different soil fractions in percentage of fallow level
Size: 90 × 8
Show 5
cc_variant | depth | Fraction | OC_ratio.m | OC_ratio.se | OC_Frac_pc | OC_Frac_pc.fal | OC_ratio.fal
Mustard | 0-10 cm | <1 | 98.3844146865413 | 3.97650226545588 | 71.7874402975763 | 69.9648186593361 | 101.8226216
Mustard | 0-10 cm | 2-1 | 86.4626899359186 | 6.62719140124324 | 10.3738651907209 | 11.4193844310946 | 98.9544807
Mustard | 0-10 cm | 4-2 | 91.4895769500228 | 2.89064327293449 | 9.74144665802457 | 10.1248819761701 | 99.6165646
Mustard | 0-10 cm | 8-4 | 80.179327313764 | 7.96623578257205 | 6.771927400681 | 8.11336815387892 | 98.6585592
Mustard | 0-10 cm | 16-8 | 132.345597976651 | 6.11664262406956 | 1.32532045299718 | 1.132640338561 | 100.1926801
Previous 1 2 3 ... 18 Next
Mustard
Clover
Oat
Phacelia
Mix4
Mix12
0-10 cm
20-30 cm
30-40 cm
OC (% of Fallow)



Implementation

```r
# R version: 4.3.2
# Requirements: tidyverse==2.0.0

library(tidyverse)

COL1 <- c("Fallow" = "black", "Mustard" = "red3", "Clover" = "OliveDrab", "Oat" = "Gold",
          "Phacelia" = "SteelBlue", "Mix4" = "orchid3", "Mix12" = "orange4")
theme_myBW <- theme(axis.title.x = element_text(size = 10, color = "black"),
                    axis.title.y = element_text(angle = 90, vjust = 1.5, size = 10, color = "black"),
                    axis.text.x = element_text(size = 10, color = "black"),
                    axis.text.y = element_text(size = 10, color = "black"),
                    axis.ticks = element_line(colour = "black"),
                    strip.text.x = element_text(size = 10, color = "black"),
                    strip.background = element_blank(),
                    panel.border = element_rect(colour = "black", fill = NA),
                    panel.grid.major = element_blank(),
                    panel.grid.minor = element_blank(),
                    plot.title = element_text(size = 12, hjust = 0.5),
                    legend.text = element_text(size = 10),
                    legend.text.align = 0,
                    legend.title = element_text(size = 10),
                    legend.key = element_rect(colour = "white", fill = "white"),
                    legend.key.size = unit(5, "mm"),
                    legend.background = element_blank(),
                    legend.position = "bottom")
se <- function(x) sqrt(var(x, na.rm = TRUE) / length(na.omit(x)))
coord_radar <- function (theta = "x", start = 0, direction = 1, clip = "off") {
  theta <- match.arg(theta, c("x", "y"))
  r <- if (theta == "x") "y" else "x"
  ggproto("CordRadar", CoordPolar, theta = theta, r = r, start = start,
          direction = sign(direction), clip = clip,
          is_linear = function(coord) TRUE)
}
df.radar <- read.csv("http://knowledgeloom21.service.tib.eu/media/files/49903279b388b50216fb92957d60d143/1nuwo314.csv")
df.radar$fraction <- factor(df.radar$fraction, levels = c("<1", "2-1", "4-2", "8-4", "16-8", "bulk"))
df.radar.m <- df.radar %>%
  filter(cc_variant != "Fallow") %>%
```

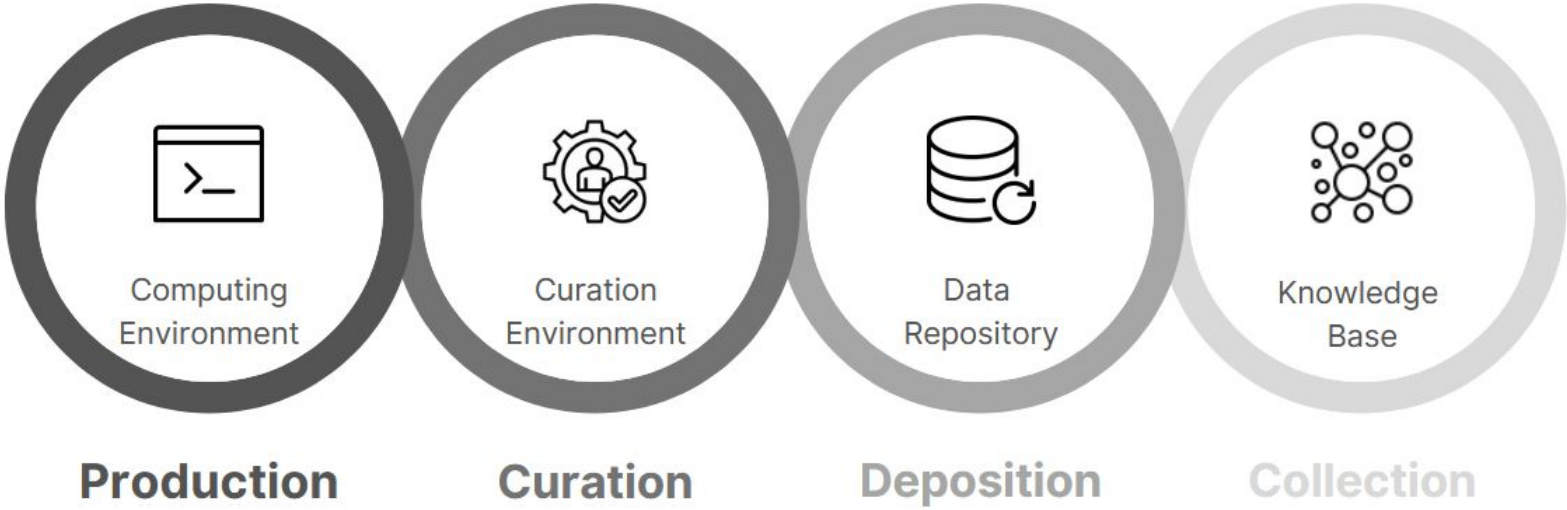
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Give it a try at knowledgeloom.tib.eu

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Questions

The Loom Record data lifecycle



Production

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```r
library(tidyverse)

COL1 <- c("Fallow" = "black", "Mustard" = "red3", "Clover" = "OliveDrab", "Oat" = "Gold",
          "Phacelia" = "SteelBlue", "Mix4" = "orchid3", "Mix12" = "orange4")
theme_myBW <- theme(axis.title.x = element_text(size = 10, color = "black"),
                    axis.title.y = element_text(angle = 90, vjust = 1.5, size = 10, color = "black"),
                    axis.text.x = element_text(size = 10, color = "black"),
                    axis.text.y = element_text(size = 10, color = "black"),
                    axis.ticks = element_line(colour = "black"),
                    strip.text.x = element_text(size = 10, color = "black"),
                    strip.background = element_blank(),
                    panel.border = element_rect(colour = "black", fill = NA),
                    panel.grid.major = element_blank(),
                    panel.grid.minor = element_blank(),
                    plot.title = element_text(size = 12, hjust = 0.5),
                    legend.text = element_text(size = 10),
                    legend.text.align = 0,
                    legend.title = element_text(size = 10),
                    legend.key = element_rect(colour = "white", fill = "white"),
                    legend.key.size = unit(5, "mm"),
                    legend.background = element_blank(),
                    legend.position = "bottom")
se <- function(x) sqrt(var(x, na.rm = TRUE) / length(na.omit(x)))
coord_radar <- function (theta = "x", start = 0, direction = 1, clip = "off") {
  theta <- match.arg(theta, c("x", "y"))
  r <- if (theta == "x") "y" else "x"
  ggproto("CordRadar", CoordPolar, theta = theta, r = r, start = start,
          direction = sign(direction), clip = clip,
          is_linear = function(coord) TRUE)
}
df.radar <- read.csv("1nuwo314.csv")
df.radar$Fraction <- factor(df.radar$Fraction, levels = c("<1", "2-1", "4-2", "8-4", "16-8", "bulk"))
df.radar.m <- df.radar %>%
  filter(cc_variant != "Fallow") %>%
  group_by(cc_variant, depth, Fraction) %>%
  summarise(OC_ratio.m = mean(OC_ratio.pc),
            OC_ratio.se = se(OC_ratio.pc),
            OC_Frac_pc = mean(OC_Frac_pc),
            OC_Frac_pc.fal = mean(OC_Frac_pc.fal),
            OC_ratio.fal = mean(OC_ratio.fal))
df.radar.F <- df.radar %>%
  filter(cc_variant != "Fallow") %>%
  group_by(cc_variant, depth, Fraction) %>%
  summarise(OC_ratio.m = 100)
ggplot(df.radar.m, aes(x = Fraction, y = OC_ratio.fal, group = cc_variant, color = cc_variant, fill = cc_variant)) +
  geom_polygon(data = df.radar.F, aes(x = Fraction, y = OC_ratio.m), color="black", fill = NA, size = 0.8, alpha = 0.8) +
  geom_polygon(size = 0.8, alpha= 0.5) +
  coord_radar(clip = "off") +
  scale_color_manual(values = COL1) +
  scale_fill_manual(values = COL1) +
  labs(y = "OC (% of Fallow)", color="", fill="") +
  facet_grid(depth~cc_variant) +
  theme_myBW +
  theme(legend.position = "bottom",
        panel.grid.major = element_line(color = "gray", size = 0.5, linetype = 2),
        axis.text.x = element_text(color = "gray12", size = 6),
        axis.title.x = element_blank(),
        axis.text.y = element_text(size = 8),
        panel.border = element_blank())
```

Scientific code

Production

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```r
library(dtreg)

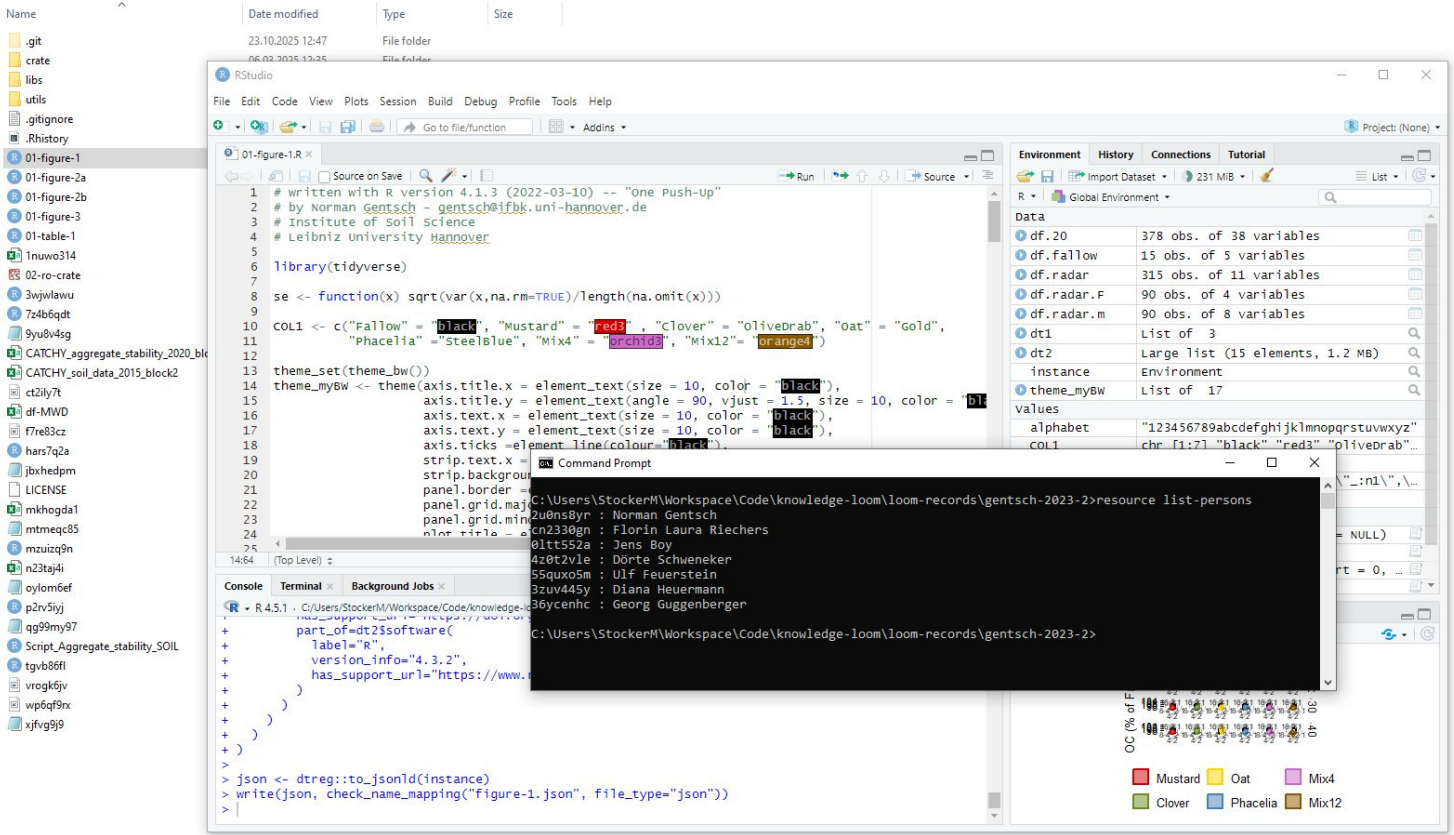
dt1 <- dtreg::load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis
dt2 <- dtreg::load_datatype("https://doi.org/21.T11969/c6b413ba96ba477b5dca") # Multilevel analysis

instance <- dt1$data_analysis(
  is_implemented_by="table-1.R",
  has_part=dt2$multilevel_analysis(
    label="Linear mixed model (LMM) fitting with mean weight diameter (MWD) ...",
    executes=dt2$software_method(
      label="lmer",
      has_support_url="https://search.r-project.org/CRAN/refmans/lmerTest/html/lmer.html",
      is_implemented_by="lmer(MWD_cor ~ cc_variant + (1|depth), data = df.MWD)",
      part_of=dt2$software_library(
        label="lmerTest"
        part_of=dt2$software(
          label="R"
        )
      )
    ),
    targets=dt2$component(label="MWD_cor"),
    level=dt2$component(label="depth"),
    has_input=dt2$data_item(
      source_url="df-mwd.csv",
      has_characteristic=dt2$matrix_size(
        number_of_rows=nrow(df.MWD)
      )
    ),
```

Loom record code

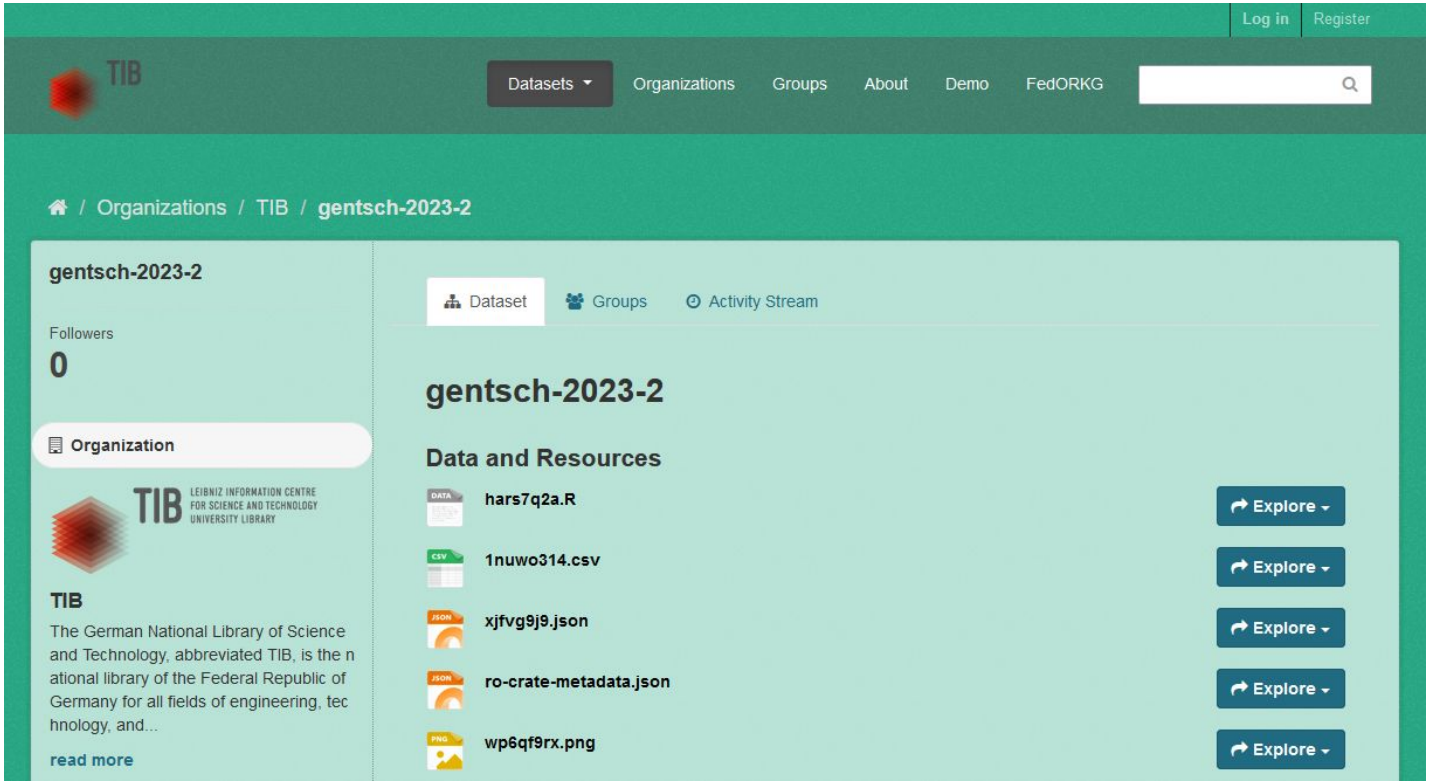
Curation

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Deposition

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Loom Record

The legacy effect of Cover Crops on soil structure amelioration

Soil Sciences · October 14, 2025 · Norman Gentsch · https://doi.org/10.48366/5eqe8313

Soil structure is highly sensitive to intensive management practices, making its amelioration a critical component of sustainable agriculture. Cover crops (CCs) are employed as part of an integrated approach to improve soil quality by stimulating plant- and microbial-mediated aggregate formation. This study investigated the legacy effect of CC applications on soil aggregation in a long-term field experiment in Germany, measuring organic carbon (OC) distribution and aggregate stability (Mean Weight Diameter, MWD) after three crop rotations in the second main crop (winter wheat) following the CC treatments. Four single CC species (mustard, phacelia, clover, and oat) and a fallow treatment were compared to a four-species mixture (Mix4) and a highly diverse 12-species mixture (Mix12). The results showed that all CC treatments significantly increased the MWD compared to the fallow, with increases ranging between 10% and 19%. Crucially, the average MWD increase was significantly higher for CC mixtures (16%) than for single CC species (12%). The highly diverse Mix12 resulted in the highest overall MWD improvement (19%). Structural Equation Modelling (SEM) demonstrated that CC type significantly impacted aggregate OC distribution. While most OC was stored in the smallest macroaggregate fraction (< 1 mm), single CCs tended to increase OC storage in this small fraction, whereas CC mixtures were more likely to increase OC accumulation in the largest macroaggregate fractions (8–16 mm), which correlate positively with MWD. Layers below the maximum cultivation depth (> 15 cm) showed the strongest positive response to CC treatments, with the biodiverse mixtures showing additional benefits in the 20–30 cm depth. These differences suggest that combining diverse functional root traits, varied litter qualities, and rhizodeposits enhance plant-microbial interactions as key mediators of aggregate formation are more effectively in mixtures than in single species. The consecutive integration of CCs provides a positive legacy effect for structure amelioration in arable soils. Increasing functional diversity in CC mixtures is recommended as a strategy to further enhance the benefits of CCs for sustainable soil management.

Statement

The variability of organic carbon distribution within soil aggregates is quite large between cover crop treatments.

Soil carbon · Soil aggregate · Cover crop

Data Analysis

Descriptive statistics

Descriptive statistics for relative proportion of organic carbon in different soil fractions in percentage of fallow level

Procedure

Executes summarise of dplyr (1.1.4) in R (4.3.2)

```
summarise(OC_ratio.m = mean(OC_ratio.pc),
  OC_ratio.se = se(OC_ratio.pc),
  OC_Frac_pc = mean(OC_Frac_pc),
  OC_Frac_pc.fal = mean(OC_Frac_pc.fal),
  OC_ratio.fal = mean(OC_ratio.fal))
```

Source Article

Cover crops improve soil structure and change organic carbon distribution in macroaggregate fractions

2024 · Norman Gentsch · Florin Laura Riechers · Jens Boy · Dörte Schweneker · Ulf Feuerstein · Diana Heuermann · Georg Guggenberger

SOIL · Copernicus Publications · https://doi.org/10.5194/soil-10-139-2024

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Producing Loom Record data
An example

The Irises example

- Test the statistical significance difference of mean petal length in Iris species
- We use the Iris flower dataset (wikipedia.org/wiki/Iris_flower_data_set)
- Classic dataset for statistical classification techniques, clustering, etc.
- Three Iris species: Setosa, Virginica, and Versicolor
- Four features: Length and width of the sepals and petals
- 50 samples from each of three species, meaning 150 samples in total
- Dataset size is thus 150 x 5 (four features plus species)
- We conduct the statistical test on Setosa and Virginica species
- We use Student's t-test as the statistical method
- We implement the statistical test in Python, using the `scipy` package

Statistical test

```
import pandas as pd
import numpy as np
from sklearn import datasets
from scipy.stats import ttest_ind

iris = datasets.load_iris()

iris = pd.DataFrame(data = np.c_[iris["data"], iris["target"]], columns = iris["feature_names"] + ["target"])

target_mapping = {0: "setosa", 1: "versicolor", 2: "virginica"}
iris["species"] = iris["target"].map(target_mapping)

iris = iris[(iris.species == "setosa") | (iris.species == "virginica")].loc[:, ["petal length (cm)", "species"]]

x = iris[iris.species == "setosa"].loc[:, "petal length (cm)"]
y = iris[iris.species == "virginica"].loc[:, "petal length (cm)"]

tt = ttest_ind(x, y, equal_var = False)

df = pd.DataFrame({ "t.statistic": [tt.statistic], "df": [tt.df], "p.value": [tt.pvalue] })
```

Loom Record data

```python
from dtreg.load_datatype import load_datatype
from dtreg.to_jsonld import to_jsonld

dt1 = load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis
dt2 = load_datatype("https://doi.org/21.T11969/b9335ce2c99ed87735a6") # Group comparison

instance = dt1.data_analysis(
  is_implemented_by="data-analysis-1.py",
  has_part=dt2.group_comparison(
    label="Group comparison with petal length dependent variable for setosa and virginica irises",
    executes=dt2.software_method(
      label="ttest_ind",
      has_support_url="https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html",
      is_implemented_by="ttest_ind(x, y, equal_var = False)",
      part_of=dt2.software_library(
        label="scipy",
        has_support_url="https://docs.scipy.org/doc/scipy/index.html",
        version_info="1.15.1",
        part_of=dt2.software(
          label="Python",
          version_info="3.12.5",
          has_support_url="https://www.python.org"
        )
      )
    ),
    targets=dt2.component(label="petal length (cm)"),
    has_input=dt2.data_item(
      label="Iris data set with species and petal length",
      source_table=iris,
      has_part=[
        dt2.component(label="petal length (cm)"),
        dt2.component(label="species")
      ],
      has_characteristic=dt2.matrix_size(
        number_of_rows=iris.shape[0],
        number_of_columns=iris.shape[1]
      )
    ),
    has_output=dt2.data_item(
      source_table=df,
      has_part=[
        dt2.component(label="t.statistic"),
        dt2.component(label="df"),
        dt2.component(label="p.value")
      ],
      has_characteristic=dt2.matrix_size(
        number_of_rows=df.shape[0],
        number_of_columns=df.shape[1]
      ),
      has_expression=dt2.figure(source_url="data-analysis-1.png")
    )
  )
)

with open("data-analysis-1.json", "w") as f:
  f.write(to_jsonld(instance))
```

Loom Record data

Part I

```
from dtreg.load_datatype import load_datatype
from dtreg.to_jsonld import to_jsonld

dt1 = load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis
dt2 = load_datatype("https://doi.org/21.T11969/b9335ce2c99ed87735a6") # Group comparison
```

Part II

```
instance = dt1.data_analysis(
  is_implemented_by="data-analysis-1.py",
  has_part=dt2.group_comparison(
    label="Group comparison with petal length dependent variable for setosa and virginica irises",
    executes=dt2.software_method(
      label="ttest_ind",
      has_support_url="https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html",
      is_implemented_by="ttest_ind(x, y, equal_var = False)",
      part_of=dt2.software_library(
        label="scipy",
        has_support_url="https://docs.scipy.org/doc/scipy/index.html",
        version_info="1.15.1",
        part_of=dt2.software(
          label="Python",
          version_info="3.12.5",
          has_support_url="https://www.python.org"
        )
      )
    ),
```

Part III

```
    targets=dt2.component(label="petal length (cm)"),
    has_input=dt2.data_item(
      label="Iris data set with species and petal length",
      source_table=iris,
      has_part=[
        dt2.component(label="petal length (cm)"),
        dt2.component(label="species")
      ],
      has_characteristic=dt2.matrix_size(
        number_of_rows=iris.shape[0],
        number_of_columns=iris.shape[1]
      )
    ),
```

Part IV

```
    has_output=dt2.data_item(
      source_table=df,
      has_part=[
        dt2.component(label="t.statistic"),
        dt2.component(label="df"),
        dt2.component(label="p.value")
      ],
      has_characteristic=dt2.matrix_size(
        number_of_rows=df.shape[0],
        number_of_columns=df.shape[1]
      ),
      has_expression=dt2.figure(source_url="data-analysis-1.png")
    )
  )
)

with open("data-analysis-1.json", "w") as f:
  f.write(to_jsonld(instance))
```

Loom Record data (Part I)

```python
from dtreg.load_datatype import load_datatype
from dtreg.to_jsonld import to_jsonld

dt1 = load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis
dt2 = load_datatype("https://doi.org/21.T11969/b9335ce2c99ed87735a6") # Group comparison
```

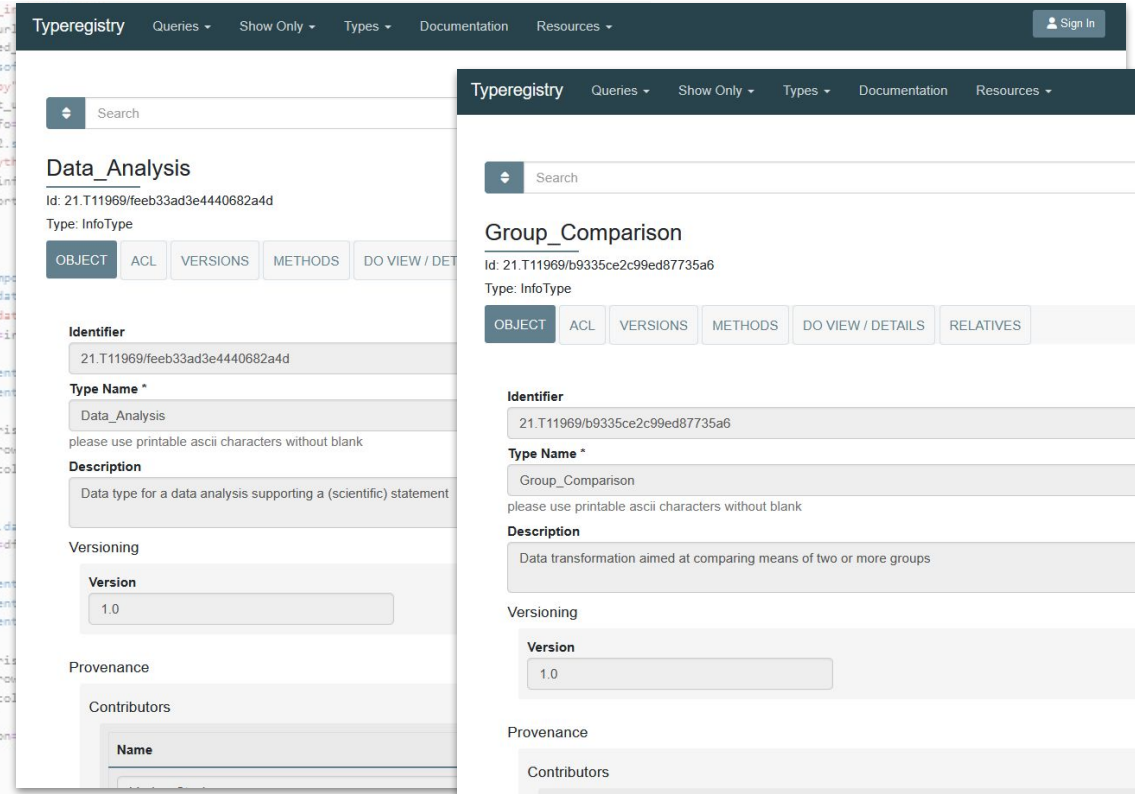
```
from dtreg.load_datatype import load_datatype
from dtreg.to_jsonld import to_jsonld

dt1 = load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis
dt2 = load_datatype("https://doi.org/21.T11969/b9335ce2c99ed87735a6") # Group comparison

instance
  is_imp
  has_pa
    labe
    exec
      label="ttest_i
      has_support_ur
      is_implemented
      part_of=dt2.so
        label="scipy
        has_support_
        version_info=
        part_of=dt2.
          label="Pyth
          version_in
          has_support
      )
    ),
    targets=dt2.comp
    has_input=dt2.dat
      label="Iris da
      source_table=ir
      has_part=[
        dt2.component
        dt2.component
      ],
      has_characteris
        number_of_ro
        number_of_co
      )
    ),
    has_output=dt2.da
      source_table=dt
      has_part=[
        dt2.component
        dt2.component
        dt2.component
      ],
      has_characteris
        number_of_ro
        number_of_co
      ),
      has_expression=
    )
  )
)

with open("data-analysis-1.json", "w") as f:
  f.write(to_jsonld(instance))
```

Loom Record data



Loom Record data (Part II)

```
instance = dt1.data_analysis(
  is_implemented_by="data-analysis-1.py",
  has_part=dt2.group_comparison(
    label="Group comparison with petal length dependent variable for setosa and virginica irises",
    executes=dt2.software_method(
      label="ttest_ind",
      has_support_url="https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html",
      is_implemented_by="ttest_ind(x, y, equal_var = False)",
      part_of=dt2.software_library(
        label="scipy",
        has_support_url="https://docs.scipy.org/doc/scipy/index.html",
        version_info="1.15.1",
        part_of=dt2.software(
          label="Python",
          version_info="3.12.5",
          has_support_url="https://www.python.org"
        )
      )
    ),
```

Loom Record data (Part III)

```
    targets=dt2.component(label="petal length (cm)"),
    has_input=dt2.data_item(
      label="Iris data set with species and petal length",
      source_table=iris,
      has_part=[
        dt2.component(label="petal length (cm)"),
        dt2.component(label="species")
      ],
      has_characteristic=dt2.matrix_size(
        number_of_rows=iris.shape[0],
        number_of_columns=iris.shape[1]
      )
    ),
```

Loom Record data (Part IV)

```
    has_output=dt2.data_item(
      source_table=df,
      has_part=[
        dt2.component(label="t.statistic"),
        dt2.component(label="df"),
        dt2.component(label="p.value")
      ],
      has_characteristic=dt2.matrix_size(
        number_of_rows=df.shape[0],
        number_of_columns=df.shape[1]
      ),
      has_expression=dt2.figure(source_url="data-analysis-1.png")
    )
  )
)

with open("data-analysis-1.json", "w") as f:
  f.write(to_jsonld(instance))
```

Loom Record data (experimental!)

```python
from mrap.analytic_instances import group_comparison
from dtreg.load_datatype import load_datatype
from dtreg.to_jsonld import to_jsonld

dt = load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis

instance = dt.data_analysis(
    is_implemented_by = "data-analysis-1.py",
    has_part = group_comparison(["scipy", "ttest_ind(x, y, equal_var = False)"], iris, df)
)

with open("data-analysis-2.json", "w") as f:
    f.write(to_jsonld(instance))
```

Loom Record data (experimental!)

```python
from mrap.analytic_instances import group_comparison
from dtreg.load_datatype import load_datatype
from dtreg.to_jsonld import to_jsonld

dt1 = load_datatype("https://doi.org/21.T11969/feeb33ad3e4440682a4d") # Data analysis
dt2 = load_datatype("https://doi.org/21.T11969/b9335ce2c99ed87735a6") # Group comparison

p1 = group_comparison(["scipy", "ttest_ind(x, y, equal_var = False)"], iris, df)

p1.label = "Group comparison with petal length dependent variable for setosa and virginica irises"
p1.executes.has_support_url = "https://docs.scipy.org/doc/scipy/reference/generated/scipy.stats.ttest_ind.html"
p1.executes.part_of.has_support_url = "https://docs.scipy.org/doc/scipy/index.html"
p1.executes.part_of.part_of.has_support_url = "https://www.python.org"
p1.targets = dt2.component(label="petal length (cm)")
p1.has_input.label = "Iris data set with species and petal length"
p1.has_input.has_part = [
    dt2.component(label="petal length (cm)"),
    dt2.component(label="species")
]
p1.has_output.has_part = [
    dt2.component(label="t.statistic"),
    dt2.component(label="df"),
    dt2.component(label="p.value")
]
p1.has_output.has_expression = dt2.figure(source_url="data-analysis-1.png")

instance = dt1.data_analysis(
    is_implemented_by = "data-analysis-1.py",
    has_part = p1
)

with open("data-analysis-2.json", "w") as f:
    f.write(to_jsonld(instance))
```

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Let’s look at this live

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Publishing Loom Record data
An example

Deposition and publication

- Preparing a Loom Record publication is somewhat involved
- However, this activity is not left to researchers
- It is done (currently) at TIB by professional data curators
- The main tasks are
  - Prepare the files to be deposited on the research data repository
  - Files include data sets, images, scripts and the JSON-LD files
  - In addition to file renaming, scripts and JSON-LD files need to be edited to read data from URLs
  - Create a RO-Crate files to describe the Loom Record data
  - Upload the files onto a research data repository, for which we use the Leibniz Data Manager

```
article = crate.add(ContextEntity(crate, "https://doi.org/10.5194/soil-10-139-2024", properties={
  "@type": [ "ScholarlyArticle" ],
  "name": "Cover crops improve soil structure and change organic carbon distribution in macroaggregate fractions",
  "abstract": "Soil structure is sensitive to intensive soil management. It can be ameliorated by a reduction in soil cultivation and stimu
  "researchField": [ {
    "@type": [ "ResearchField" ],
    "label": "Soil Science",
    "relatedIdentifier": [
        "https://orkg.org/field/R153"
    ]
  } ],
  "author": [ gentsch ],
  "identifier": "https://doi.org/10.5194/soil-10-139-2024",
  "publisher": {
    "@type": [ "Publisher" ],
    "label": "Copernicus Publications"
  },
  "journal": {
    "@type": [ "Journal" ],
    "label": "SOIL"
  },
  "datePublished": "2024"
}))

crate = addToCrate(crate, "figure-1.json", properties={
  "@type": [ "File" ],
  "encodingFormat": "application/ld+json",
  "version": "1.0",
  "author": [ gentsch ],
  "supports": [ crate.add(ContextEntity(crate, None, properties={
    "@type": [ "Statement" ],
    "version": "1.0",
    "author": [ gentsch ],
    "notation": crate.add(ContextEntity(crate, None, properties={
      "@type": [ "LinguisticStatement" ],
      "label": "The variability of organic carbon distribution within soil aggregates is quite large between cover crop treatments",
      "concept": [ concept1 ]
    }))
  })) ],
  "components": [ component1, component2 ]
})
```

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Questions

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Discussion

Discussion topics

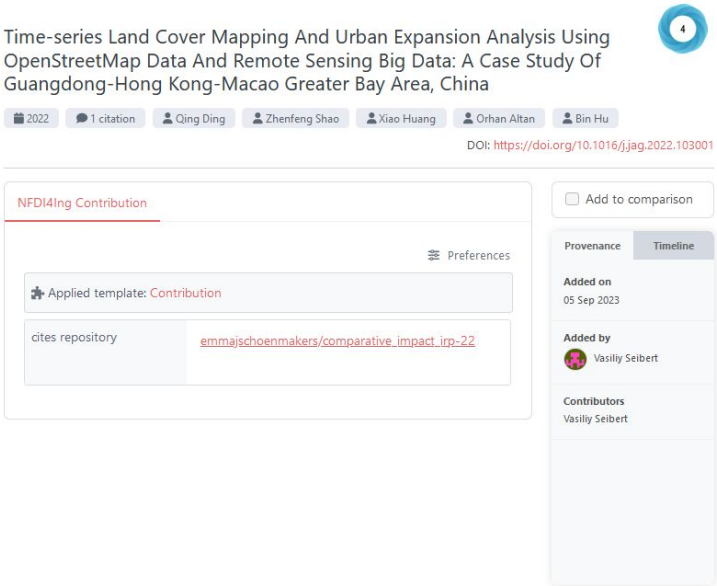
- Comparing approaches for structured knowledge production
- Data quality
- Reproducibility of findings
- Interlinking and discovery of artefacts

Comparing techniques

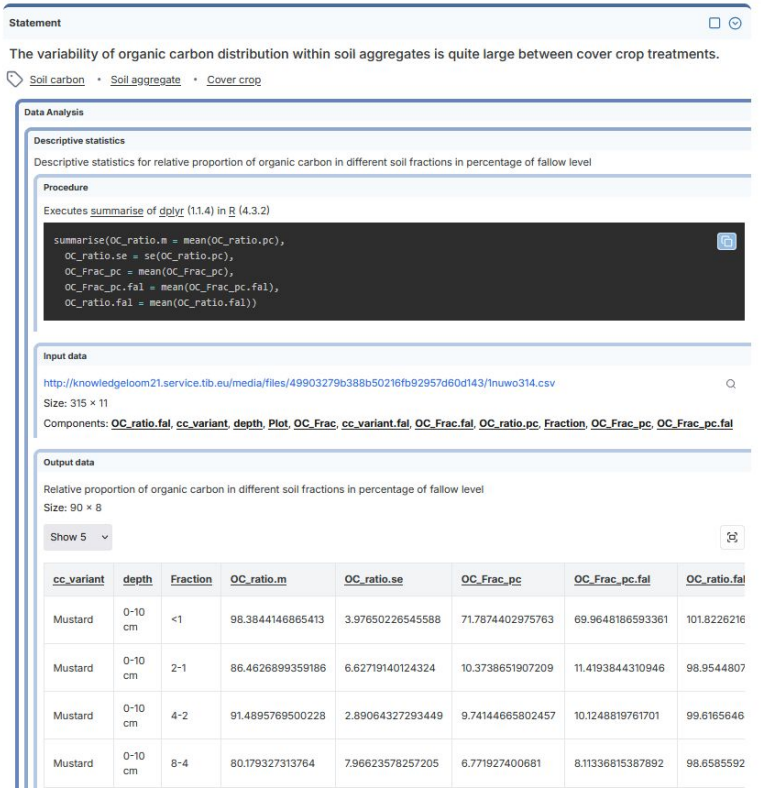
	Accuracy	Richness	Coverage	Simplicity (*)	Legacy	Scalability
Manual	✅	❌	✅(1)	✅	✅	❌
Semi-automated	❌	❌❌	✅✅	❌❌	✅✅	✅✅
Born-reusable	✅✅	✅✅	✅(1)	✅✅	❌(2)	❌

(1) Assuming broad participation
(2) Assuming the availability of data and code, the approach is in principle applicable also to legacy content
(*) Technological simplicity

Data quality



vs



Reproducibility

Cover crops improve soil structure and change organic carbon distribution in macroaggregate fractions
Norman Gentsch et al.

3.2 Aggregate distribution and OC in aggregate fractions

Most of the soil aggregates (57.5 % to 74.2 %) were stored in the <1 mm fraction, highest in the topsoil and decreasing with soil depth (Fig. S6 in the Supplement). The intermediate fractions (8–4, 4–2, 2–1 mm) ranged from 5.6 % to 13.8 %, each. The largest fraction (16–8 mm) contributed least to the total soil aggregates (0 % to 7.0 %). A similar distribution was found for the proportion of OC in different aggregate fractions (Fig. S7 in the Supplement). The variability of OC distribution within aggregates was quite large between (Fig. 1) but also within CC treatments (Fig. S8). Thus, only a few of the observed trends were significantly different by pairwise comparison of treatments (Fig. S8). For example, the largest differences among CC treatments were found in the top 0–10 cm, between oat and Mix12, Mix4, and fallow. Of all treatments, oat led to the highest proportion of OC in the <1 mm fraction (Fig. 1), which was a significant difference from Mix12 (p=0.038) in the uppermost layer (Fig. S8). All pure cultivated CCs tended to store more OC in the <1 mm fraction in the 0–10 cm increment, whereas mixtures stored a greater proportion in the 16–8 mm fraction. In the soil increments below 20 cm, the data variability was large between treatments. Here, only Mix4 led to approximately 1 % more OC in the 2–1 mm fraction than oat at the 30–40 cm depth (Fig. S8).

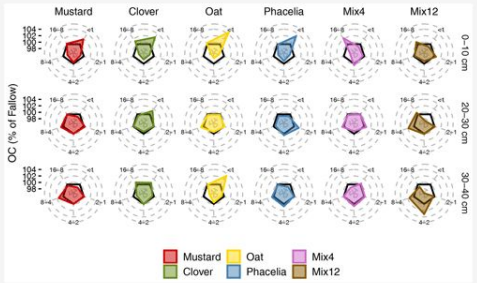


Figure 1 Relative proportion of OC in different soil fractions relative to the fallow level. Fallow as 100 % is marked by the black polygon in the background. Polygons represent the means of three replicates, and statistical evaluation of the graph is shown in Fig. S8.

▸ Download

VS

Statement

The variability of organic carbon distribution within soil aggregates is quite large between cover crop treatments.

Soil carbon • Soil aggregate • Cover crop

Data Analysis

Descriptive statistics

Implementation

```r
# R version: 4.3.2
# Requirements: tidyverse==2.0.0

library(tidyverse)

COL1 <- c("Fallow" = "black", "Mustard" = "red3", "Clover" = "OliveDrab", "Oat" = "Gold",
          "Phacelia" = "SteelBlue", "Mix4" = "orchid3", "Mix12" = "orange4")
theme_myBW <- theme(axis.title.x = element_text(size = 10, color = "black"),
                    axis.title.y = element_text(angle = 90, vjust = 1.5, size = 10, color = "black"),
                    axis.text.x = element_text(size = 10, color = "black"),
                    axis.text.y = element_text(size = 10, color = "black"),
                    axis.ticks = element_line(colour = "black"),
                    strip.text.x = element_text(size = 10, color = "black"),
                    strip.background = element_blank(),
                    panel.border = element_rect(colour = "black", fill = NA),
                    panel.grid.major = element_blank(),
                    panel.grid.minor = element_blank(),
                    plot.title = element_text(size = 12, hjust = 0.5),
                    legend.text = element_text(size = 10),
                    legend.text.align = 0,
                    legend.title = element_text(size = 10),
                    legend.key = element_rect(colour = "white", fill = "white"),
                    legend.key.size = unit(5, "mm"),
                    legend.background = element_blank(),
                    legend.position = "bottom")
se <- function(x) sqrt(var(x, na.rm = TRUE) / length(na.omit(x)))
coord_radar <- function (theta = "x", start = 0, direction = 1, clip = "off") {
  theta <- match.arg(theta, c("x", "y"))
  r <- if (theta == "x") "y" else "x"
  ggproto("CordRadar", CoordPolar, theta = theta, r = r, start = start,
          direction = sign(direction), clip = clip,
          is_linear = function(coord) TRUE)
}
df.radar <- read.csv("http://knowledgeloom21.service.tib.eu/media/files/49903279b388b50216fb92957d60d143/1nuwo314.csv")
df.radar$Fraction <- factor(df.radar$Fraction, levels = c("<1", "2-1", "4-2", "8-4", "16-8", "bulk"))
df.radar.m <- df.radar %>%
  filter(cc_variant != "Fallow") %>%
```

Interlinking and discovery (I)



Interlinking and discovery (II)




```
data:
  id: "10.48366/r664252"
  type: "dois"
  attributes:
    doi: "10.48366/r664252"
    prefix: "10.48366"
    suffix: "r664252"
    identifiers: []
    alternateIdentifiers: []
    creators:
      0:
        name: "Norman Gentsch"
        nameType: "Personal"
        nameIdentifiers:
          0:
            schemeUri: "https://orcid.org"
            nameIdentifier: "https://orcid.org/0000-0003-1166-8973"
            nameIdentifierScheme: "ORCID"
        affiliation: []
    titles:
      ...ibution in macroaggregate fractions"
      ris: "DATA"
      bibtex: "misc"
      citeproc: "dataset"
      schemaOrg: "Dataset"
      resourceType: "Paper"
      resourceTypeGeneral: "Dataset"
    relatedIdentifiers:
      0:
        relationType: "IsVariantFormOf"
        relatedIdentifier: "https://doi.org/10.5194/soil-10-139-2024"
        relatedIdentifierType: "DOI"
    relatedItems: []
    sizes: []
    formats: []
```

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Questions

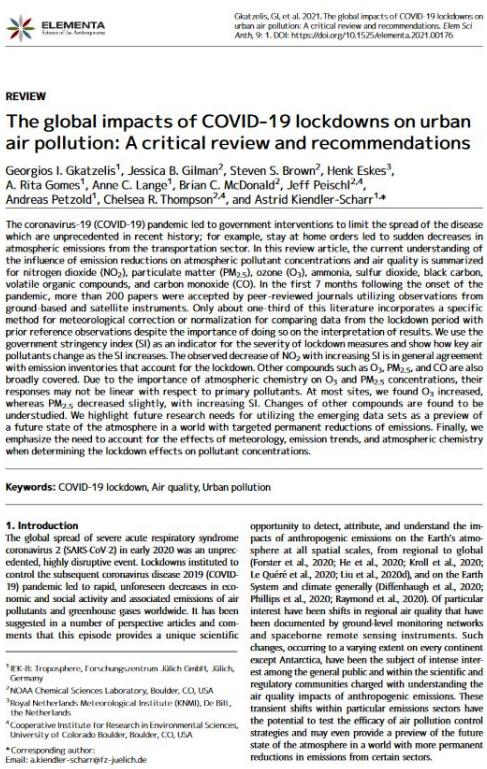
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Supporting synthesis research

Why do we do what we do?

- Access, clarity, reproducibility, peer-review, archiving, etc. are all goals
- The primary goal is to support synthesis research
- Meaning, integrate and analyze statement data from numerous articles
- To support meta-analysis, systematic reviews, or other synthesis types
- The rationale being to
  - Reduce reliance on extraction
  - Increase accuracy and efficiency
  - Lowering infrastructure “friction”

Synthesis research: An example (I)



ELEMENTA

Gkatzelis, GI, et al. 2021. The global impacts of COVID-19 lockdowns on urban air pollution: A critical review and recommendations. Elem Sci Anth, 9: 1. DOI: https://doi.org/10.1525/elementa.2021.00176

REVIEW

The global impacts of COVID-19 lockdowns on urban air pollution: A critical review and recommendations

Georgios I. Gkatzelis[1], Jessica B. Gilman[2], Steven S. Brown[2], Henk Eskes[3], A. Rita Gomes[1], Anne C. Lange[1], Brian C. McDonald[2], Jeff Peischl[2,4], Andreas Petzold[1], Chelsea R. Thompson[2,4], and Astrid Kiendler-Scharr[1]*

The coronavirus-19 (COVID-19) pandemic led to government interventions to limit the spread of the disease which are unprecedented in recent history; for example, stay at home orders led to sudden decreases in atmospheric emissions from the transportation sector. In this review article, the current understanding of the influence of emission reductions on atmospheric pollutant concentrations and air quality is summarized for nitrogen dioxide (NO_2), particulate matter ($PM_{2.5}$), ozone (O_3), ammonia, sulfur dioxide, black carbon, volatile organic compounds, and carbon monoxide (CO). In the first 7 months following the onset of the pandemic, more than 200 papers were accepted by peer-reviewed journals utilizing observations from ground-based and satellite instruments. Only about one-third of this literature incorporates a specific method for meteorological correction or normalization for comparing data from the lockdown period with prior reference observations despite the importance of doing so on the interpretation of results. We use the government stringency index (SI) as an indicator for the severity of lockdown measures and show how key air pollutants change as the SI increases. The observed decrease of NO_2 with increasing SI is in general agreement with emission inventories that account for the lockdown. Other compounds such as O_3, $PM_{2.5}$, and CO are also broadly covered. Due to the importance of atmospheric chemistry on O_3 and $PM_{2.5}$ concentrations, their responses may not be linear with respect to primary pollutants. At most sites, we found O_3 increased, whereas $PM_{2.5}$ decreased slightly, with increasing SI. Changes of other compounds are found to be understudied. We highlight future research needs for utilizing the emerging data sets as a preview of a future state of the atmosphere in a world with targeted permanent reductions of emissions. Finally, we emphasize the need to account for the effects of meteorology, emission trends, and atmospheric chemistry when determining the lockdown effects on pollutant concentrations.

Keywords: COVID-19 lockdown, Air quality, Urban pollution

1. Introduction

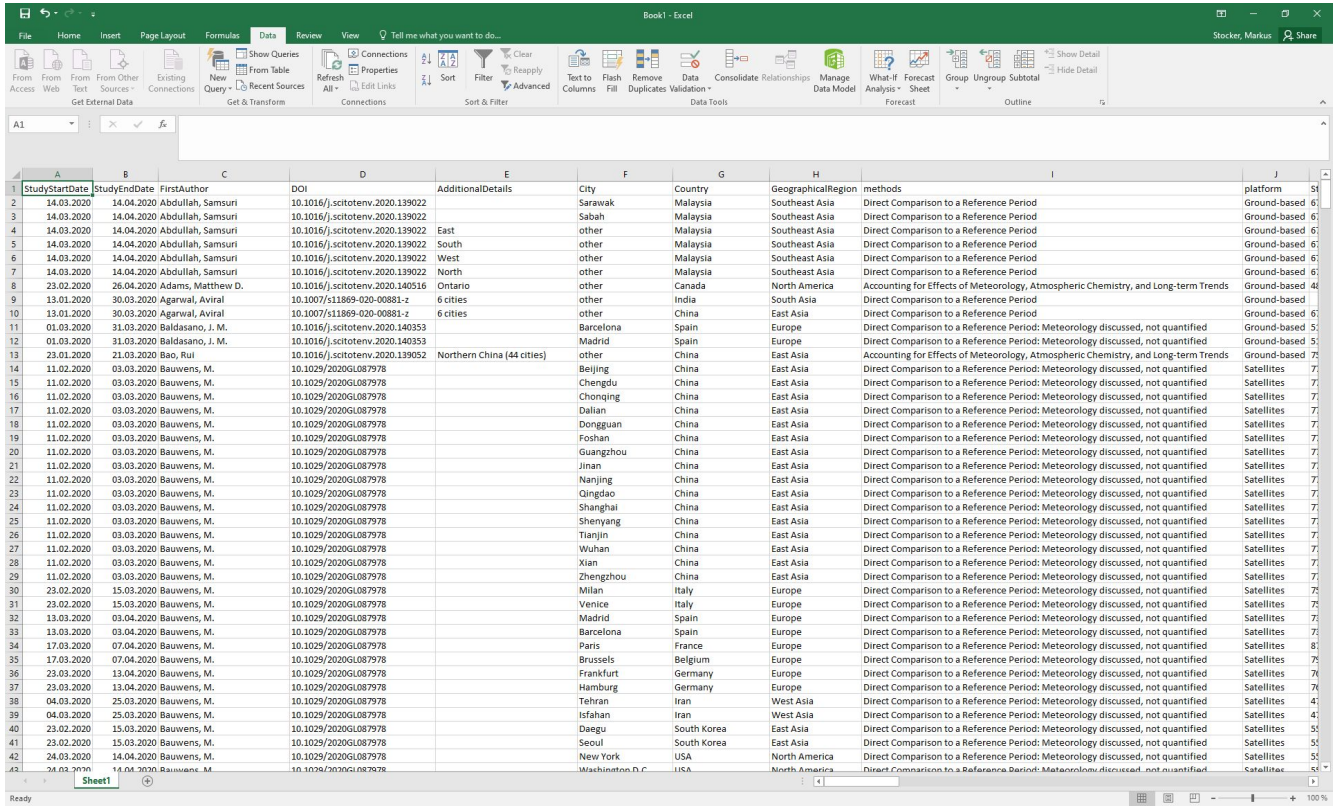
The global spread of severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2) in early 2020 was an unprecedented, highly disruptive event. Lockdowns instituted to control the subsequent coronavirus disease 2019 (COVID-19) pandemic led to rapid, unforeseen decreases in economic and social activity and associated emissions of air pollutants and greenhouse gases worldwide. It has been suggested in a number of perspective articles and comments that this episode provides a unique scientific opportunity to detect, attribute, and understand the impacts of anthropogenic emissions on the Earth's atmosphere at all spatial scales, from regional to global (Forster et al., 2020; He et al., 2020; Kroll et al., 2020; Le Quéré et al., 2020; Liu et al., 2020d), and on the Earth System and climate generally (Diffenbaugh et al., 2020; Phillips et al., 2020; Raymond et al., 2020). Of particular interest have been shifts in regional air quality that have been documented by ground-level monitoring networks and spaceborne remote sensing instruments. Such changes, occurring to a varying extent on every continent except Antarctica, have been the subject of intense interest among the general public and within the scientific and regulatory communities charged with understanding the air quality impacts of anthropogenic emissions. These transient shifts within particular emissions sectors have the potential to test the efficacy of air pollution control strategies and may even provide a preview of the future state of the atmosphere in a world with more permanent reductions in emissions from certain sectors.

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[2] NOAA Chemical Sciences Laboratory, Boulder, CO, USA
[3] Royal Netherlands Meteorological Institute (KNMI), De Bilt, the Netherlands
[4] Cooperative Institute for Research in Environmental Sciences, University of Colorado Boulder, Boulder, CO, USA
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https://doi.org/10.1525/elementa.2021.00176

Synthesis research: An example (I)



Book1 - Excel

StudyStartDate	StudyEndDate	FirstAuthor	DOI	AdditionalDetails	City	Country	GeographicalRegion	methods	platform
14.03.2020	14.04.2020	Abdullah, Samsuri	10.1016/j.scitotenv.2020.139022		Sarawak	Malaysia	Southeast Asia	Direct Comparison to a Reference Period	Ground-based
14.03.2020	14.04.2020	Abdullah, Samsuri	10.1016/j.scitotenv.2020.139022		Sabah	Malaysia	Southeast Asia	Direct Comparison to a Reference Period	Ground-based
14.03.2020	14.04.2020	Abdullah, Samsuri	10.1016/j.scitotenv.2020.139022	East	other	Malaysia	Southeast Asia	Direct Comparison to a Reference Period	Ground-based
14.03.2020	14.04.2020	Abdullah, Samsuri	10.1016/j.scitotenv.2020.139022	South	other	Malaysia	Southeast Asia	Direct Comparison to a Reference Period	Ground-based
14.03.2020	14.04.2020	Abdullah, Samsuri	10.1016/j.scitotenv.2020.139022	West	other	Malaysia	Southeast Asia	Direct Comparison to a Reference Period	Ground-based
14.03.2020	14.04.2020	Abdullah, Samsuri	10.1016/j.scitotenv.2020.139022	North	other	Malaysia	Southeast Asia	Direct Comparison to a Reference Period	Ground-based
23.02.2020	26.04.2020	Adams, Matthew D.	10.1016/j.scitotenv.2020.140516	Ontario	other	Canada	North America	Accounting for Effects of Meteorology, Atmospheric Chemistry, and Long-term Trends	Ground-based
13.01.2020	30.03.2020	Agarwal, Aviral	10.1007/s11869-020-00881-z	6 cities	other	India	South Asia	Direct Comparison to a Reference Period	Ground-based
13.01.2020	30.03.2020	Agarwal, Aviral	10.1007/s11869-020-00881-z	6 cities	other	China	East Asia	Direct Comparison to a Reference Period	Ground-based
01.03.2020	31.03.2020	Baldasano, J. M.	10.1016/j.scitotenv.2020.140353		Barcelona	Spain	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Ground-based
01.03.2020	31.03.2020	Baldasano, J. M.	10.1016/j.scitotenv.2020.140353		Madrid	Spain	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Ground-based
23.01.2020	21.03.2020	Bao, Rui	10.1016/j.scitotenv.2020.139052	Northern China (44 cities)	other	China	East Asia	Accounting for Effects of Meteorology, Atmospheric Chemistry, and Long-term Trends	Ground-based
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Beijing	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Chengdu	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Chongqing	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Dalian	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Dongguan	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Foshan	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Guangzhou	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Jinan	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Nanjing	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Qingdao	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Shanghai	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Shenyang	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Tianjin	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Wuhan	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Xian	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
11.02.2020	03.03.2020	Bauwens, M.	10.1029/2020GL087978		Zhengzhou	China	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
23.02.2020	15.03.2020	Bauwens, M.	10.1029/2020GL087978		Milan	Italy	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
23.02.2020	15.03.2020	Bauwens, M.	10.1029/2020GL087978		Venice	Italy	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
13.03.2020	03.04.2020	Bauwens, M.	10.1029/2020GL087978		Madrid	Spain	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
13.03.2020	03.04.2020	Bauwens, M.	10.1029/2020GL087978		Barcelona	Spain	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
17.03.2020	07.04.2020	Bauwens, M.	10.1029/2020GL087978		Paris	France	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
17.03.2020	07.04.2020	Bauwens, M.	10.1029/2020GL087978		Brussels	Belgium	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
23.03.2020	13.04.2020	Bauwens, M.	10.1029/2020GL087978		Frankfurt	Germany	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
23.03.2020	13.04.2020	Bauwens, M.	10.1029/2020GL087978		Hamburg	Germany	Europe	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
04.03.2020	25.03.2020	Bauwens, M.	10.1029/2020GL087978		Tehran	Iran	West Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
04.03.2020	25.03.2020	Bauwens, M.	10.1029/2020GL087978		Isfahan	Iran	West Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
23.02.2020	15.03.2020	Bauwens, M.	10.1029/2020GL087978		Daegu	South Korea	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
23.02.2020	15.03.2020	Bauwens, M.	10.1029/2020GL087978		Seoul	South Korea	East Asia	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
24.03.2020	14.04.2020	Bauwens, M.	10.1029/2020GL087978		New York	USA	North America	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites
24.03.2020	14.04.2020	Bauwens, M.	10.1029/2020GL087978		Washington D.C.	USA	North America	Direct Comparison to a Reference Period: Meteorology discussed, not quantified	Satellites

Synthesis research with the TIB Knowledge Loom

» Based on model results, there was a significant negative relationship between the Shannon diversity index and B. oleae abundance

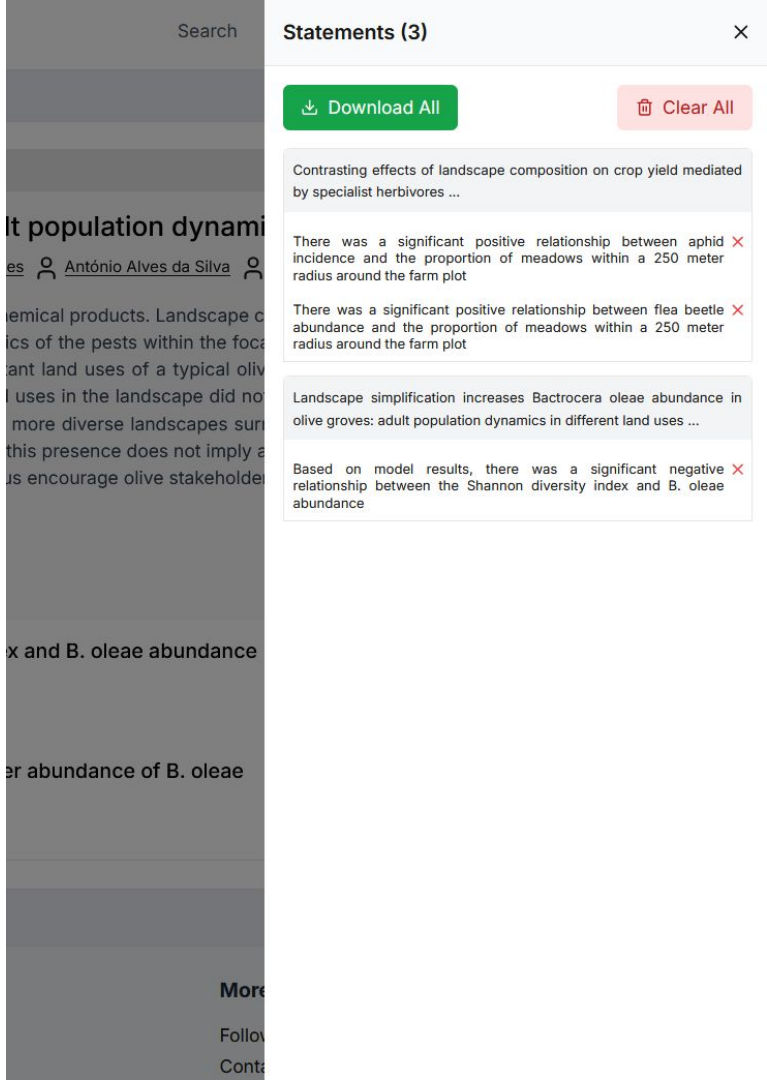
Daniel Paredes

» There was a significant positive relationship between aphid incidence and the proportion of meadows within a 250 meter radius around the farm plot

Ricardo Perez-Alvarez

» There was a significant positive relationship between flea beetle abundance and the proportion of meadows within a 250 meter radius around the farm plot

Ricardo Perez-Alvarez



Synthesis research with the TIB Knowledge Loom

» Based on model results, there was a significant negative relatio
between the Shannon diversity index and B. oleae abundance

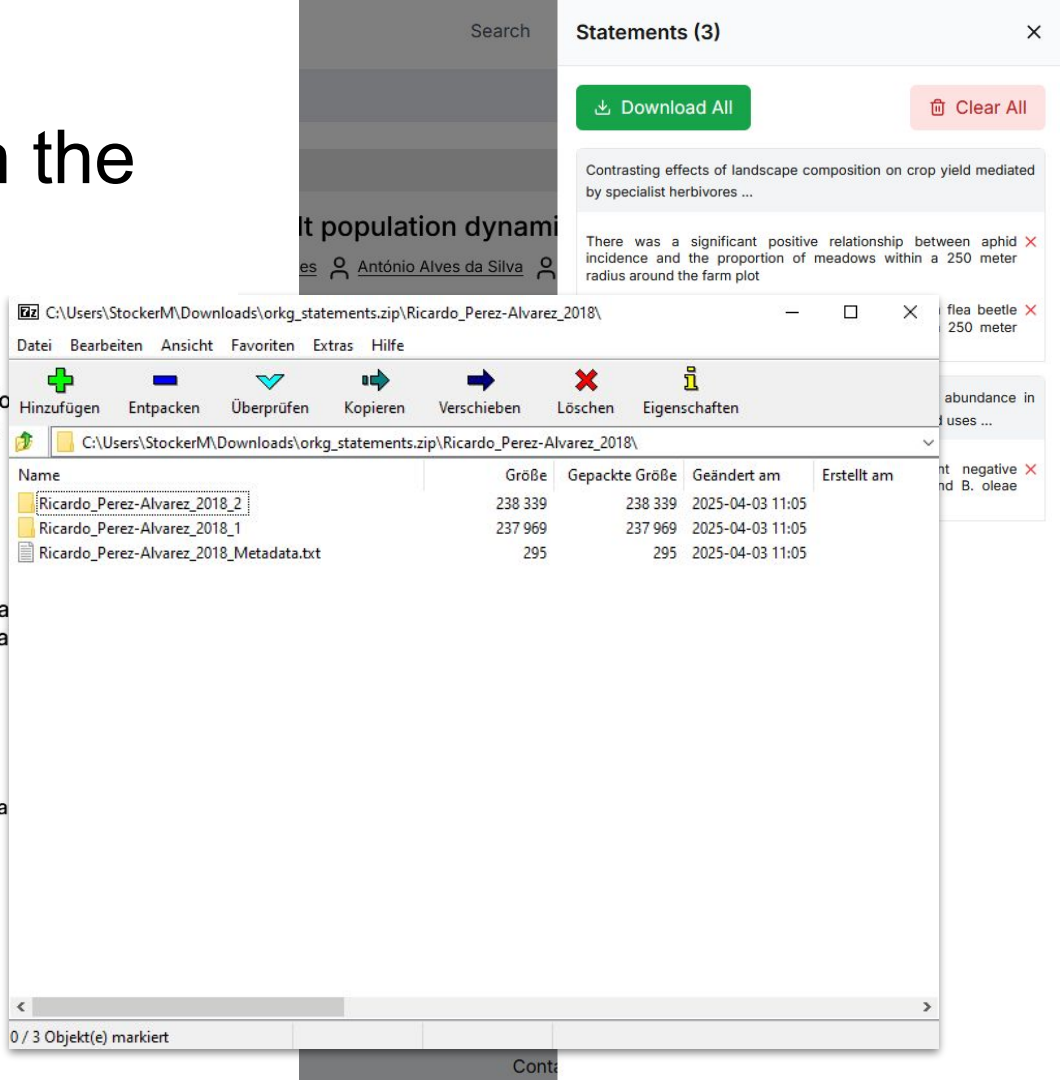
Daniel Paredes

» There was a significant positive relationship between a
the proportion of meadows within a 250 meter radius a

Ricardo Perez-Alvarez

» There was a significant positive relationship between flea beetle abunda
proportion of meadows within a 250 meter radius around the farm plot

Ricardo Perez-Alvarez



Synthesis research with the TIB Knowledge Loom

```r
# Retrieve significance testing output dataset for Ricardo's fig. 4a (aphid incidence) - here we need the p-value
# https://reborn.orkg.org/statement/ef47c5fb81675d030a25569c4f8309a6
perez_fig_4a <- read.csv(
  "data/Ricardo_Perez-Alvarez_2018/Ricardo_Perez-Alvarez_2018_1/Ricardo_Perez-Alvarez_2018_1_3/Ricardo_Perez-Alvarez_2018_1_3_Output_Data_1.csv",
  row.names=1
)
# Retrieve significance testing output dataset for Ricardo's fig. 4b (flea beetle abundance) - here we need the p-value
# https://reborn.orkg.org/statement/ee80591c4b6278ba6605f923a4256b9c
perez_fig_4b <- read.csv(
  "data/Ricardo_Perez-Alvarez_2018/Ricardo_Perez-Alvarez_2018_2/Ricardo_Perez-Alvarez_2018_2_3/Ricardo_Perez-Alvarez_2018_2_3_Output_Data_1.csv",
  row.names=1
)
# Retrieve significance testing output dataset for Ricardo's fig. 4c (incidence of lepidopteran larvae) - here we need the p-value
# https://reborn.orkg.org/statement/b50c409710fb6558c80a860653126bba
perez_fig_4c <- read.csv(
  "data/Ricardo_Perez-Alvarez_2018/Ricardo_Perez-Alvarez_2018_3/Ricardo_Perez-Alvarez_2018_3_3/Ricardo_Perez-Alvarez_2018_3_3_Output_Data_1.csv",
  row.names=1
)

# Retrieve ANOVA output dataset for Daniel's fig. 4 (Bactrocera oleae abundance) (n=25) - here we want the Pr..Chi value
# https://reborn.orkg.org/statement/34401eca0b28c95b7122b7fd9e9c4027
data <- read.csv(
  "data/Daniel_Paredes_2022/Daniel_Paredes_2022_1/Daniel_Paredes_2022_1_1/Daniel_Paredes_2022_1_1_Input_Data_1.csv"
)

glm.bo.shdi <- glm.nb(bo~shdi,data=data)
paredes_fig_4_anova <- data.frame(anova(glm.bo.shdi))

...
```

Try it out with https://gitlab.com/TIBHannover/orkg/orkg-reborn/use-case-projects/mini-synthesis-ecology

Takeaways

- Digitalization is well-underway also in research and scholarly communication
- Infrastructures have standardized **metadata**, identification, access
- For numerous assets (articles, data, code, samples, instruments, etc.)
- As well as for contributors (researchers) and organizations
- The potential of modern knowledge management has not been reached
- It is possible to demonstrate this potential, but realization remains a challenge
- AI technologies enable new things, but still fail to realize the potential

Leibniz Universität Hannover

Questions

Next steps (I)

- Let’s apply what we learned for concrete articles
- How do we access articles, data, and code?
- Obtain articles in progress or already published by researchers at your faculty
- Get in touch with doctoral or postdoctoral students
- Ask for copies of articles, data, code, and support with an occasional question
- The research work should employ Python or R to produce results
- In computer science, for instance benchmark-based algorithm evaluation
- For instance, performance of machine learning models
- In other fields, say economics, likely statistical data analysis
- Alternatively, we draw from our own pool of candidate articles
- Work individually or together with a fellow student

Next steps (II)

- Use our curation board on Gitlab to select articles and organize the work
- We organize the collected materials (articles, data, code) in Gitlab issues
- Assign yourself to issues for the articles you decided to work on
- In the table you'll find in the Stud.IP Wiki
  - Write your Gitlab.com account ID
  - Write the issue identifiers ("name-year-1-1") of articles you processed

Next steps (III)

- If not done, you may want to start with the Iris example
  - https://knowledgeloom.tib.eu/pages/help
- To start, candidate articles can be found here
  - https://gitlab.com/groups/TIBHannover/lki/knowledge-loom/loom-records/-/boards
  - Panel “Laboratory Students”
- Possibly, reach out to teaching staff at the faculty to obtain other articles
- Check out https://huggingface.co/papers/trending to discover articles with code
  - If you manage to execute the code, they are likely good candidates